

An Intelligent IoT and AI-Based Maternal Health Monitoring, Risk Prediction and Emergency Alert System

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Board Declaration

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Declaration

We, **Takia Yasmin, and Joya Rani Saha**, hereby declare that the research work titled "*An Intelligent IoT and AI-Based Maternal Health Monitoring, Risk Prediction and Emergency Alert System*" is our original work. This research has been conducted as a partial fulfillment of the requirements for the degree of **Bachelor of Science in Internet of Things (IoT) and Robotics Engineering(IRE)** at **University of Frontier Technology, Bangladesh (UFTB)**.

We affirm that this work has not been submitted previously for any degree, diploma, or other qualifications at any other institution or university. All references, ideas, and contributions from others used in this research are duly cited and acknowledged.

This research was conducted under the supervision and guidance of **Mostafiz Ahammed, Lecturer of Department of IRE**. We sincerely acknowledge his invaluable insights and support in shaping this work.

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Abstract

Materna AI is a system that uses Internet of Things and Artificial Intelligence to keep track of the health and safety of women especially those in rural and low-resource areas. Materna AI uses an ESP32 microcontroller with ECG, MAX30100, MPU6050, BMP280 and GPS modules to gather data about body signals, motion, environment and location. The ESP32 reads the sensors. Sends the data over Wi-Fi to an AI backend. To predict health risk for mothers Materna AI uses age, blood pressure, blood sugar, body temperature and heart rate. Materna AI tried six machine learning models: Logistic Regression, Decision Tree, K-Nearest Neighbors, Gradient Boosting, XGBoost and Random Forest. Random Forest gave the accuracy of 93.08 percent and Materna AI chose it to split risk into low, medium and high groups. For fall detection Materna AI used a set of 6,446 pictures to compare four YOLO models – YOLOv8n, YOLOv9t, YOLO11s and YOLO26n. Materna AI chose YOLOv8n for use because it had a recall of 99.7 percent, a mean average precision of 99.4 percent at 50 percent overlap and it could process each picture in only 2.3 milliseconds. Materna AI also has a 128×64 SSD1306 OLED screen that shows heart rate, SpO₂, ECG value, temperature, pressure and the system status. Materna AI turns on an ECG buzzer when the raw ECG value is than 500 or more than 3000. Materna AI sends ECG data and GPS location, over Bluetooth. It sends other data to ThingSpeak for cloud monitoring. Materna AI also offers a Flask web dashboard that helps manage patients and doctors set appointments, track history and watch GPS from afar. By putting together sensors, machine learning, fall detection, ECG alerts, Bluetooth, GPS, cloud services and a web dashboard Materna AI gives a cheap, growable and multi-purpose system that watches maternal health in real time and offers emergency help.

Keywords: Maternal Health Monitoring, IoT, ESP32, ECG, SpO₂, Random Forest, YOLO, Fall Detection, GPS, ThingSpeak, Flask.

Acronyms List

- AI : Artificial Intelligence
- IoT : Internet of Things
- ML : Machine Learning
- ECG : Electrocardiogram
- SpO2 : Oxygen Saturation
- BPM : Beats Per Minute
- GPS : Global Positioning System
- LCD : Liquid Crystal Display
- WiFi : Wireless Fidelity
- ESP32 : Espressif 32-bit Microcontroller
- MPU6050 : Motion Processing Unit Sensor
- HR : Heart Rate
- HTTP : Hypertext Transfer Protocol
- KNN : K-Nearest Neighbors
- YOLO : You Only Look Once
- IoU : Intersection over Union
- mAP : Mean Average Precision
- API : Application Programming Interface

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Chapter 1

Introduction

1.1 Background

Maternal healthcare is a public health concern, especially in developing countries where access to regular medical supervision and emergency healthcare services can be limited. Pregnancy is a period during which the physical and physiological conditions of a mother can change rapidly. Several complications, such as heart rate, high blood pressure, diabetes, abnormal body temperature and other health-related problems may occur during pregnancy. If these conditions are not detected and managed in time they can create risks for both the mother and the baby. Therefore continuous monitoring and early identification of health conditions are essential for improving maternal safety.

Traditional maternal healthcare mainly depends on hospital visits manual measurement of health parameters and direct communication between patients and healthcare professionals[8]. Although these methods are useful they may not provide monitoring. Many patients, those living in rural or remote areas may face difficulties in visiting hospitals frequently because of distance, transportation problems, financial limitations or a shortage of medical professionals. As a result important changes in a patient's health condition may remain unnoticed until the situation becomes serious. This creates a need for an remote healthcare monitoring system that can collect health data continuously and provide timely information to doctors and caregivers[27].

IoT-based healthcare systems can collect data using sensors and transmit the information to a monitoring platform through wireless communication[4]. In a health monitoring system sensors can be used to measure parameters such as heart rate, oxygen saturation, body temperature, movement and environmental conditions[11,14]. An ESP32 microcontroller can act as the central processing unit by collecting data from sensors[1]. This approach can reduce the need for physical checkups and allow healthcare professionals to monitor patients from a distance.

In addition to sensor-based monitoring machine learning can support the identification of health risks from collected patient information[13]. By analyzing parameters such as age, blood pressure, blood sugar, temperature and heart rate, machine learning models can classify patients into risk categories. Such a system may help doctors identify patients

who require observation or additional medical consultation.

Patient safety is another concern during pregnancy. A fall abnormal movement or emergency condition may require immediate assistance. To address this issue an accelerometer and gyroscope-based MPU6050 sensor can be used for fall detection[7]. A GPS module can provide the patient's location during an emergency while Bluetooth communication can transmit information to a nearby device. In addition an ECG-based buzzer alert can provide a warning when the measured raw ECG value crosses a predefined threshold. These features can improve the ability of the system to identify emergencies and support faster responses.

Local and remote data visualization is also necessary for healthcare monitoring. A 128×64 SSD1306 OLED display can show sensor readings and system status on the hardware device. At the time Wi-Fi communication can transmit the collected data to a Flask-based web platform and the ThingSpeak cloud service. The web-based dashboard can provide doctors with information real-time health parameters, medical history, consultation facilities and GPS location tracking.

Considering these requirements the proposed Materna AI system combines sensors an ESP32 controller, machine learning fall detection, GPS tracking, Bluetooth communication, ECG-based alerts and a web-based dashboard into a single healthcare framework[9,10]. The system is designed to support maternal health monitoring, early risk identification, emergency assistance and communication between patients and healthcare providers[33]. By integrating hardware-based sensing with data analysis and remote monitoring Materna AI aims to provide a more accessible, responsive and technology-supported approach, to maternal healthcare.

1.2 Motivation

Maternal healthcare matters a lot because complications, during pregnancy can appear quickly and put the mother and baby in danger. Even though regular medical visits are important it is hard to keep watch all the time. This is because there are not health centres treatments cost a lot hospitals are far away and there are not enough doctors. For

these reasons we need an remote monitoring system. This system can watch health, spot possible risks and give quick information to doctors.

Furthermore several challenges persist in healthcare monitoring:

1. **Lack of Continuous Monitoring and Emergency Support:** Traditional healthcare systems rely on scheduled hospital visits and manual health measurements. Traditional healthcare systems miss shifts, in heart rate, temperature, oxygen saturation or body movement. Traditional healthcare systems cannot always catch a fall or a crisis that needs instant help. Traditional healthcare systems demand real-time monitoring, fall detection, GPS tracking and automatic alert systems.
2. **Limited Remote Healthcare Access and Data Management:** Patients who live in remote places often find it hard to see doctors and go to hospitals on a regular basis. Patients also sometimes do not get their health data in an organized way. That is why we need a system that uses IoT sensors machine learning, for risk prediction Bluetooth and Wi-Fi and a web-based dashboard. This single system will help with patient monitoring keeping medical history giving online consultation and sharing emergency locations. This single system will also improve patient monitoring and make the web-based dashboard easier to use.

1.3 Objective

The objectives of this study are as follows:

1. To build an ESP32-based IoT system that monitors maternal health signs in real time such, as ECG, heart rate, SpO₂, temperature and body movement. The system will transmit data continuously to provide timely feedback and support for maternal care.
2. To implement a system that uses machine learning to predict the risk of health problems during pregnancy. The goal is to sort patients into three groups: those who're at low risk those who are at medium risk and those who are, at high risk.

3. To integrate fall detection, GPS tracking, ECG-based buzzer alerts, Bluetooth communication and an OLED display so that patient safety is better and emergency help is faster.
4. To d a website that doctors can use to keep track of their patients. This website will use Wi-Fi and a program called Flask. The website will let doctors see information, health data past medical records, doctor visits and where the patient is located using GPS. It will all work from a distance.

1.4 Contributions

The main contributions of this paper are as follows:

1. We develop an ESP32-based IoT system for real-time maternal health monitoring using ECG, MAX30100, MPU6050, BMP280, GPS, and OLED display modules.
2. We integrate machine learning-based maternal health risk prediction with fall detection, ECG-based buzzer alerts, Bluetooth communication, and GPS tracking to improve patient safety and emergency support.
3. We develop a web-based healthcare dashboard using Wi-Fi and Flask that enables doctors to monitor patient information, health parameters, medical history, consultations, and GPS location remotely.

1.5 Organization of the Thesis

The Thesis is organized as follows:

- **Chapter 1: Introduction**

This chapter provides the background, motivation, objectives, and contributions of the study.

- **Chapter 2: Literature Survey**

This chapter discusses existing research on IoT-based maternal health monitoring,

machine learning-based maternal health risk prediction, wearable healthcare systems, and fall detection techniques.

- **Chapter 3: Material and Methods**

This chapter describes the hardware components, software tools, development environment, and overall system implementation. It also explains the ESP32 controller, sensors, OLED display, Bluetooth module, buzzer, GPS module, Flask framework, ThingSpeak platform, and web-based healthcare dashboard.

- **Chapter 4: The Proposed Framework**

This chapter presents the machine learning pipeline, dataset description, data pre-processing, selected machine learning algorithms, model selection, deployment process, and performance evaluation. It also discusses the development and deployment of the human fall detection model.

- **Chapter 5: Results**

This chapter presents the experimental setup, hardware implementation, Bluetooth communication results, fall detection performance, and overall system evaluation. The chapter also discusses the effectiveness and limitations of the proposed Materna AI system.

- **Chapter 6: Conclusion**

This chapter summarizes the main contributions, key findings, and performance metrics of the study. It also discusses the limitations of the system and presents possible directions for future research.

Chapter 2

Literature Survey

2.1 Introduction

To detect health risk at an early stage monitoring the physiological conditions in maternal health care is important. Conventional assessments of maternal health are generally based on periodic clinical measurements, clinical assessment, and manual evaluation. This may not be adequate for continuous monitoring outside of clinical settings. With the advent of IoT, ML and computer vision, various intelligent healthcare systems have been proposed for real-time monitoring, risk prediction and emergency detection.

Research works have been carried out on IoT-based maternal health monitoring system, machine learning-based maternal risk prediction, wearable sensing technologies, and intelligent healthcare platform. Meanwhile, researchers have also explored computer vision and deep learning techniques for real-time human fall detection. These advances will enable the link-up of multiple monitoring and AI-based components in one maternal healthcare system.

A literature review on monitoring maternal health, machine learning-based maternal risk prediction, Internet of Things (IoT) based health care systems and vision-based fall detection is presented in this chapter. The review studies are evaluated based on their methodologies, sensory technologies, machine learning models, datasets, and system capabilities. Ultimately, the research gap targeted by the Materna AI system is identified from the drawbacks of existing approaches.

2.2 Existing Work

2.2.1 IoT-Based Maternal Health Monitoring

In recent years, IoT technology has been widely used in healthcare system to realize continuous collection and remote monitoring of physiological information. Several works have proposed IoT based systems for monitoring maternal health parameters and sending the collected information to remote healthcare platforms.

Ahmed and Kashem [6] proposed an IoT based maternal health monitoring approach for prediction of maternal health risk. Their work demonstrated the potential for combin-

ing physiological monitoring and IoT communication for supporting maternal healthcare, particularly for Bangladesh. This approach can allow for remote monitoring of health information and may reduce the need for frequent in-person visits.

Karventhan et al. [2] proposed a wearable monitoring method for ongoing surveillance of mother and fetus. The system is designed to gather physiological data from pregnant women and send the measurements for analysis. Similarly, other works have explored IoT-based healthcare architectures incorporating sensors, wireless communication, cloud platforms, and alerting systems to enable remote maternal monitoring [5], [13], [14].

Overall, studies suggest that IoT can be an effective framework for ongoing surveillance of maternal health and for remote healthcare services. However, many existing systems focus on physiological data collection and remote visualization rather than the integration of continuous sensing, machine learning-based maternal risk prediction and emergency fall detection.

2.2.2 Machine Learning-Based Maternal Health Risk Prediction

Machine learning techniques have been widely exploited for maternal health risk prediction based on physiological and clinical parameters. The methods can classify pregnant women into different risk levels according to their health information.

Malde et al. [13] investigated machine learning methods for maternal health risk prediction based on physiological and clinical parameters. Their work demonstrated the potential of machine learning for differentiating different maternal risk levels.

Simegn and Degu [24] proposed a machine learning based approach for pregnancy risk prediction and demonstrated the use of classification models for identifying different maternal risk levels.

Assaduzzaman et al. [33] evaluated multiple machine learning algorithms, including the Random Forest, Decision Tree, XGBoost, and Gradient Boosting, for maternal health risk prediction based on the Maternal Health Risk dataset. Their work demonstrated the effectiveness of machine learning for maternal risk classification.

Tappari et al. [1] integrated IoT-based monitoring with Random Forest for maternal

health risk assessment, demonstrating the potential of real-time monitoring and machine learning based risk prediction.

Overall, the works demonstrated that machine learning can effectively support maternal health risk prediction, but there is still room for an integrated system that combines real-time IoT based monitoring with machine learning based risk prediction and emergency detection.

2.2.3 Wearable and Sensor Based Healthcare Systems

Wearable and sensor based healthcare systems have been widely explored for continuous maternal health monitoring. The systems can acquire important physiological parameters including the heart rate, blood pressure, body temperature, blood oxygen level, etc.

Karventhan et al. [2] proposed a wearable system for continuous monitoring of maternal and fetal health based on physiological parameters. Priyanka et al. [17] developed an IoT-based system for pregnancy monitoring that employs sensors for acquiring maternal health data and providing alerts when needed. Similarly, Yenurkar et al. [8] investigated an integrated IoT based healthcare approach that employs sensors, communication modules, cloud platforms, and alert mechanisms for remote health monitoring. The approaches demonstrate that wearable sensors and IoT can effectively support continuous maternal health monitoring, but sensor based monitoring alone cannot be sufficient for detecting emergency situations like falls. Thus, the integration of physiological sensors and intelligent risk prediction and fall detection can enable more effective maternal healthcare.

2.2.4 Vision-Based Human Fall Detection

Computer vision has been widely explored for detecting human falls based on images or videos captured by cameras. The approaches can effectively identify abnormal body movements and provide timely alerts in case of emergency.

Harrou et al. [7] investigated a vision based approach for human fall detection based on information captured by cameras. Their work demonstrated the potential of computer vision for automatic fall detection.

Seflek and Barstuğan [5] investigated different YOLO based models for human fall detection. Their work demonstrated that YOLO-based approaches can provide effective and accurate fall detection.

Hu et al. [10] proposed a YOLO-based approach for fall detection that employs edge AI for real-time detection. Gomes et al. [6] provided a survey on computer vision based fall detection methods, demonstrating the increasing use of deep learning approaches for automatic fall detection.

Overall, the works demonstrated that computer vision and YOLO based approaches can provide effective fall detection solutions, but most of them only focus on fall detection, without considering integration with maternal health monitoring and emergency risk prediction.

Table 2.1: Comparison of Relevant Previous Studies and the Proposed Materna System

Study	Model Method /	IoT / Data	Main Contribution	Limitation / Research Gap
Tappari et al. [1]	Random Forest, IoT	Wearable sensors, ESP32, ThingSpeak	Maternal and child monitoring	No fall detection, GPS, or doctor interaction
Karventhan et al. [2]	IoT wearable	Wearable belt, cloud	Maternal and fetal monitoring	No ML risk prediction or fall detection
Seflek and Barstuğan [5]	YOLO, Radar	Micro-Doppler radar	Fall detection	Radar-based; no maternal healthcare
Ahmed and Kashem [9]	ML, IoT	Maternal health data	Maternal risk prediction	No fall detection, GPS, or doctor interaction
Hu et al. [10]	YOLO, Edge AI	60-GHz mmWave radar	Vital monitoring and fall detection	Radar-based; no complete maternal platform
Malde et al. [13]	Machine Learning	Maternal health parameters	Maternal risk prediction	No real-time IoT or fall detection
Priyanka et al. [17]	IoT monitoring	BP, HR, temperature	Pregnancy vital monitoring	No ML prediction or fall detection
Ettiyan and Geetha [18]	IoT, AI	IoT sensors, maternal/fetal data	Maternal and fetal monitoring	No integrated risk prediction, GPS, or fall detection
Simegn and Degu [24]	Machine Learning	Clinical data, web interface	Pregnancy risk classification	No real-time IoT or fall detection
Hemanth and Aishwarya [27]	IoT, ML	IoT sensors, web platform	Monitoring, prediction, fall detection	Limited GPS, cloud, and doctor services
Assaduzzaman et al. [33]	RF, DT, XG-Boost, GB	UCI maternal risk dataset	Maternal risk prediction	Static dataset; no IoT or fall detection
Proposed System	Random Forest, YOLOv8n	ESP32, ECG, MAX30105, BMP280, MPU6050, GPS, Bluetooth, Wi-Fi, ThingSpeak	Monitoring, risk prediction, fall detection, GPS, alerts, doctor interaction	Needs clinical testing, calibration, and real-world validation

The comparison, in Table 2.1 shows that existing studies have addressed aspects of ma-

ternal healthcare. These aspects include IoT-based monitoring, machine learning-based risk prediction, wearable sensing and fall detection. In cases these aspects are treated as separate modules. The proposed Materna system brings all of these aspects together in one platform by integrating real-time monitoring Random Forest-based maternal risk prediction, YOLOv8n-based fall detection, GPS tracking, emergency alerts, cloud monitoring and doctor-patient interaction and materna therefore offers a more unified way to keep track of maternal health continuously.

Chapter 3

Materials and Methods

3.1 Introduction

In this chapter, the materials, technologies and methodologies employed in the design and development of the proposed Materna AI system are presented. The system proposed in this research integrates IoT based sensing, AI, machine learning, computer vision and cloud monitoring, with the aim to enable continuous monitoring of maternal health and emergency detection. The chapter describes elaborately the hardware components, the software tools, the sensor interfacing, the data acquisition, the communications and the AI-based analysis methods that were employed in the system.

The hardware section explains the ESP32 microcontroller and the sensors attached to it that will gather physiological data, environmental data, motion data, and location data. The software section covers the development environment, backend server, machine learning models, YOLO based fall detection framework, cloud monitoring platform and web based healthcare platform. The approach used to do maternal health risk prediction, ECG abnormality detection and vision-based fall detection is also illustrated.

Moreover, this chapter describes the process of transmitting the collected sensor data to the backend server using Wi-Fi, and how it is processed and analyzed with AI. The integration of the hardware, software, AI models, cloud platform and web interface is described to illustrate the full flow of the proposed system. Finally, the overall system architecture and operational workflow are described, giving an overall picture of how the different system components can cooperate to achieve the real-time monitoring of maternal health and support for maternal safety.

3.2 Hardware Components

The proposed Materna AI system will comprise multiple hardware elements that will be designed to gather, analyse and send real-time health and safety data.

Table 3.1: Hardware Components Used in the Proposed System

SL	Hardware Component	Purpose
1	ESP32 Microcontroller	Main controller and Wi-Fi communication
2	ECG Sensor	ECG signal monitoring
3	MAX30100	Heart rate and SpO ₂ monitoring
4	BMP280	Temperature and pressure monitoring
5	MPU6050	Motion and acceleration monitoring
6	GPS Module	Location tracking
7	LCD Display	Local data visualization
8	Bluetooth Module	Short-range wireless communication
9	Buzzer	Audible emergency alert
10	Camera	Vision-based fall detection

3.2.1 ESP32 Microcontroller

The ESP32 is the main microcontroller used in the proposed Materna AI system. It is the main communication and control facility, which interfaces the various sensors and peripheral devices. The ECG sensor, MAX30100 pulse oximeter, BMP280, MPU6050, and GPS module are all utilized by the ESP32. It does some initial data processing and sends data collected via WiFi to the Flask-based backend server and to monitoring application in the cloud. The ESP32 also interfaces with the LCD display and emergency notification elements, which enable local monitoring and emergency notification.



Figure 3.1: ESP32 microcontroller used in the proposed Materna AI system.

3.2.2 ECG Sensor

The sensor used to capture heart's electrical activity is called ECG sensor. The proposed Materna AI system involves the sensor being installed in the mother to continuously monitor her heart via ECG. The collected ECG data is then sent to the Wi-Fi-enabled ESP32 microcontroller which forwards the data to Flask-based backend for further analysis. This module is useful in monitoring cardiac activity and to aid in the diagnosis of abnormal ECG conditions.

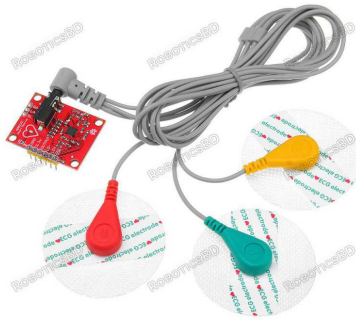


Figure 3.2: ECG Sensor used for maternal cardiac monitoring.

3.2.3 MAX30105 Pulse Oximeter Sensor

The MAX30105 sensor is used in the proposed Materna AI system for physiological monitoring. It supplies optical sensing data which can be employed to track heart-related parameters and blood oxygenation. The readings from the sensor are captured by the ESP32 microcontroller and sent via Wi-Fi to the back-end server to process or monitor further. This type of sensor can be used for continuous monitoring of maternal physiological conditions.



Figure 3.3: MAX30105 sensor used for physiological monitoring.

3.2.4 BMP280 Sensor

The proposed Materna AI system is equipped with the BMP280 sensor measuring temperature and atmospheric pressure. The other sensor readings are gathered by the ESP32 microcontroller together with these environmental readings. The collected data is wirelessly sent to the backend server and cloud monitoring platform for further processing, visualization. The BMP280, therefore, helps in the environmental and physiological condition monitoring of the proposed system..



Figure 3.4: BMP280 sensor used for temperature and pressure monitoring.

3.2.5 MPU6050 Motion Sensor

The MPU6050 is a motion sensing device to sense the acceleration and movement of the monitored mother. It can give X, Y, and Z axis acceleration data, which can be used to analyze changes in body motion and motion circumstances. The proposed Materna AI system involves the MPU6050 connecting to the ESP32 microcontroller to gather and analyze the motion data. All of the gathered data supports system overall safety and fall monitoring system.

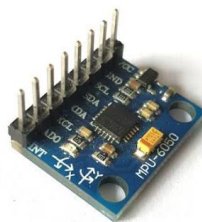


Figure 3.5: MPU6050 motion sensor used for movement monitoring.

3.2.6 GPS Module

The GPS module is utilized for the proposed Materna AI system for the current location of monitored mother. It offers users geographical details including latitude and longitude that can be used to establish the user's location when in emergency situations. ESP32 will communicate with the GPS module and the location data obtained can be sent over the IoT communication network. This feature can be helpful in emergency response situations, as it will give caregivers or responsible persons location information.



Figure 3.6: GPS module used for real-time location tracking.

3.2.7 LCD Display

The proposed Materna AI system utilizes the LCD display for local visualization of health and environmental data collected. It offers a user-friendly front end that allows for the direct display of selected sensor readings from the monitoring device. The information on the parameters, like heart rate, SpO₂ measurement, temperature, and other related information of the system, may be displayed based on the system configuration. The LCD is connected to the ESP32 microcontroller for local observation of monitoring information.



Figure 3.7: LCD display used for local visualization of monitoring data.

3.2.8 Bluetooth Module

The Bluetooth module is used in the proposed Materna AI system to provide short-range wireless communication between the monitoring device and nearby compatible devices. It can also facilitate communication and data transfer on the local level when necessary. The Bluetooth module is connected to the ESP32 microcontroller, and the data sent to it are only a subset of the data being monitored; they can be sent wirelessly without a direct wired connection.



Figure 3.8: Bluetooth module used for short-range wireless communication.

3.2.9 Buzzer and Alert Unit

A buzzer is used as a local warning element in proposed Materna AI system. It gives audible alarm when an emergency or critical condition is sensed. The sensor data is used in conjunction with the system decision making process to trigger the alert mechanism, which is directed by the ESP32 microcontroller. This feature aids to attract the attention of the monitored mother or other close by caregivers in situations that may be dangerous.



Figure 3.9: Buzzer used for local emergency alerts.

3.3 Hardware Integration and Interfacing

The proposed Materna AI system hardware is built around the ESP32 microcontroller which will serve as the main data acquisition and communication hardware. The ECG sensor, MAX30105, BMP280, MPU6050, GPS module, LCD, Bluetooth module and buzzer are connected to the ESP32 to execute their functions. The ESP32 gathers the sensor data, carries out preliminary processing and sends the necessary information via wireless to the back end server and monitoring platform in the cloud.

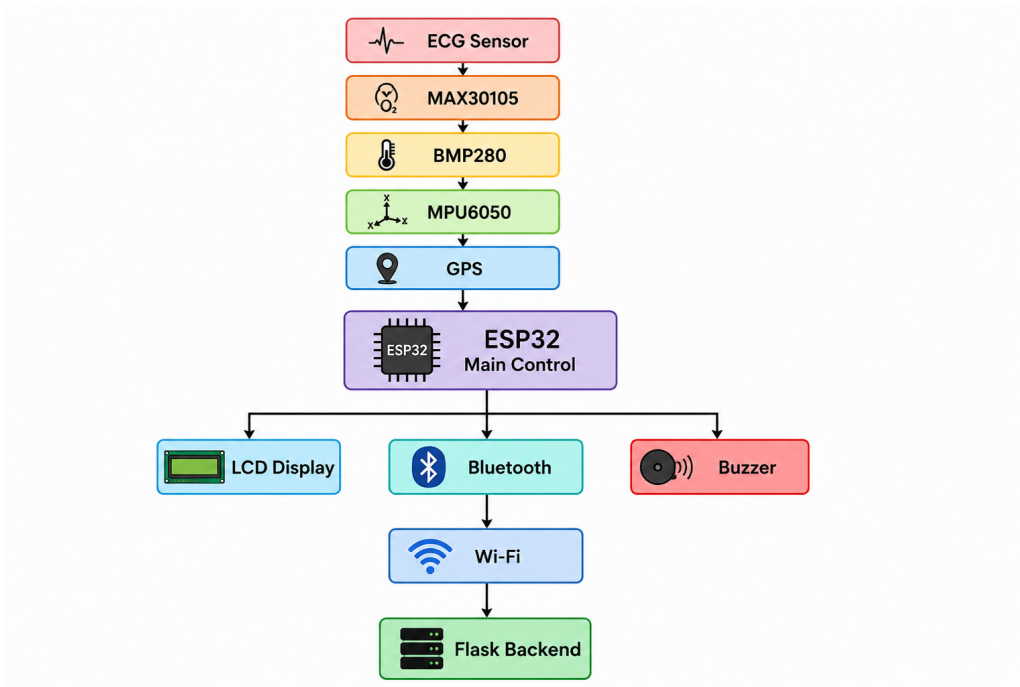


Figure 3.10: Hardware integration and interfacing of the proposed Materna AI system.

3.4 Software Components and Development Environment

The proposed Materna AI system includes multiple software elements for sensor data acquisition, data processing, maternal health risk prediction, fall detection, database management, and remote monitoring. software architecture consists of embedded programming, backend server with Flask, machine learning models, computer vision, and a web-based monitoring interface. All of these elements are designed to cooperate and

analyze the gathered information and deliver real-time health supervision and emergency help.

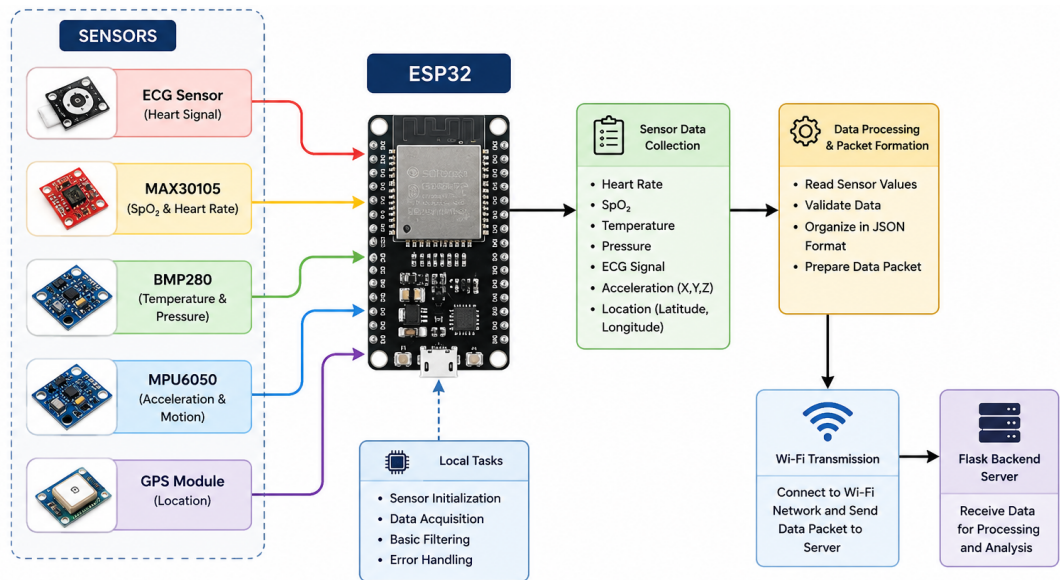


Figure 3.11: ESP32 Programming and Data Acquisition Flow of the Proposed Materna AI System

3.4.1 Development Environment – Visual Studio Code

The development environment to implement and integrate the Materna AI system has been developed primarily with Visual Studio Code (VS Code). This environment has been used to integrate and test flask backend, web interface, database related scripts and trained machine learning models. In addition, the integrated terminal is utilized to start and monitor the Flask application, during development. The VS Code environment with the developed application and running flask server is displayed in figure 3.12.

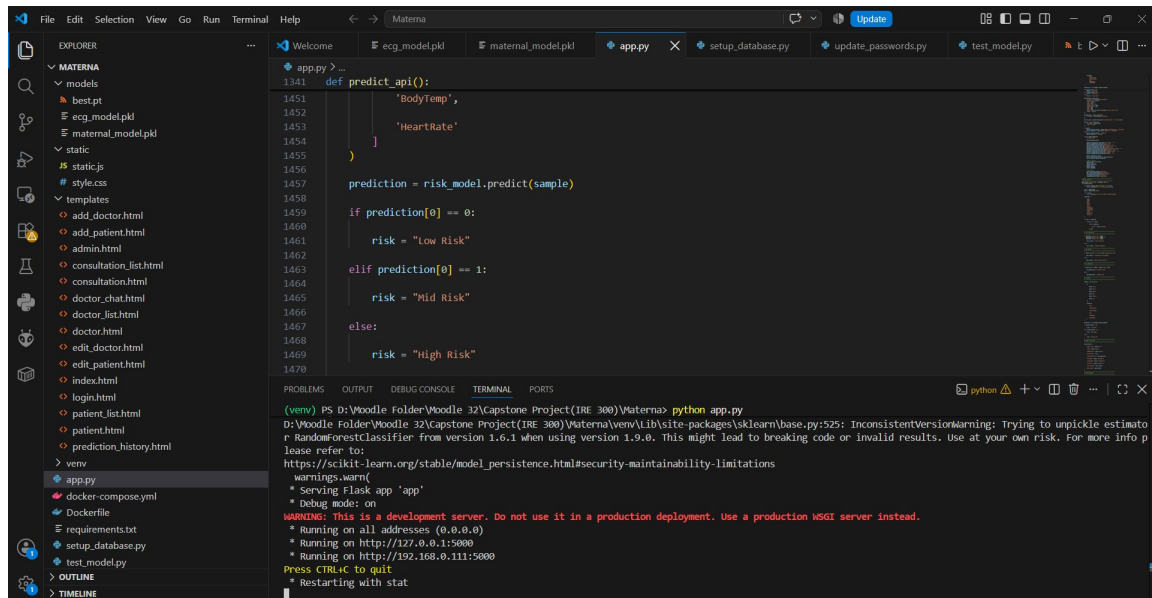


Figure 3.12: Visual Studio Code development environment and running Flask server

3.4.2 Python Programming Language

Python was used as the main programming language for developing the software components of the Materna AI system. It was employed for the development of the back-end, processing of sensor data, integration of machine learning models, database usage, and communication between the various parts of the system. In addition, Python offers a number of libraries that can be used for web development, data processing, and machine learning applications, which is suitable for implementing the proposed system.

3.4.3 Flask Web Framework

Flask was used as the backend web framework of the Materna AI system. It handles the communication among the web interface, database, and integrated machine learning models. The Flask application will be used to process prediction requests, and retrieves stored information, and provides the results to the web interface. It also facilitates the creation of APIs needed for the communication between the various modules of the system.

3.4.4 MongoDB Database

MongoDB is used as the database management system for the proposed Materna AI platform. It stores and manages the system data required for patient monitoring and web-based services. The database is used to maintain patient information, health predictions, consultation records, chat messages, and system logs. The Flask backend communicates with MongoDB to store new data and retrieve existing records when required.

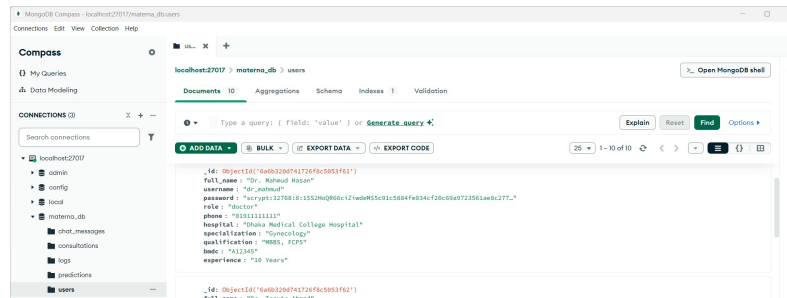


Figure 3.13: MongoDB database and collections used in the Materna AI system

3.4.5 Maternal Health Risk Prediction Model Integration

The trained maternal health risk prediction model is now part of the Flask backend of the proposed system. When a prediction is run the trained maternal health risk prediction model receives health parameters such, as body temperature and heart rate. The trained maternal health risk prediction model then processes these input values. Classifies the maternal health condition into three risk levels: Low Risk, Mid Risk and High Risk. The trained maternal health risk prediction model sends the predicted result to the web interface and stores it in the database for reference.

3.4.6 Web-Based Monitoring Interface

A web-based monitoring interface was created to offer a way to check maternal health information and system results. The interface lets people who are allowed to use it see details about patients, health data, risk predictions, advice given and if a fall has been detected. It also gives access, to parts of the system through a clear dashboard, which helps in keeping track of the information that is gathered and in taking care of it.

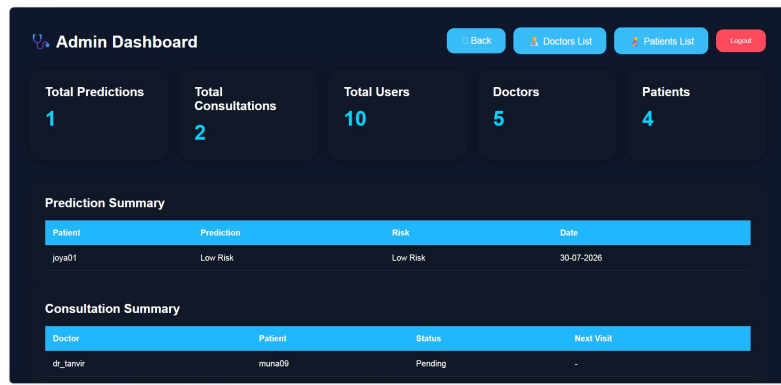


Figure 3.14: Web-based monitoring dashboard of the Materna AI system

3.4.7 ThingSpeak Cloud Platform

ThingSpeak is used in the Materna AI system as a cloud-based platform, for monitoring and showing sensor data. The sensor readings that are collected by the ESP32 are sent to the ThingSpeak channel. There the data are shown using separate field charts. This makes it possible to watch the monitored parameters over time from a distance.

In the system that was created different sensor parameters are put into fields of the ThingSpeak channel. These include ECG, Xaxis acceleration, Y-axis acceleration, Z-axis acceleration, temperature, pressure, latitude and longitude. The way these parameters are shown in graphs helps to see how the data changes and helps with monitoring of the maternal health monitoring system. We can see the ThingSpeak cloud dashboard, with the sensor data fields in Figure 3.15.

3.5 Web-Based Healthcare Platform

The Materna AI system that is being suggested has a web-based healthcare platform. This platform offers functions for patients, doctors and administrators. The platform works with the Flask backend and the MongoDB database. This helps to handle information, health tracking data AI prediction results, records of consultations and emergency situations. Each user role gets an interface based on what they need to do. The platform also has features like predicting risks, for health analyzing ECG detecting falls, tracking



Figure 3.15: ThingSpeak cloud platform showing sensor data visualization

with GPS allowing doctor consultations and sending emergency alerts.

3.5.1 User Login and Authentication

The web platform offers a login system that lets users access the application safely. Users type in their credentials and the login system checks them based on the role they have. When the login is successful the login system sends the user to the dashboard. The login system shows the dashboard, for administrator, doctor or patient access.

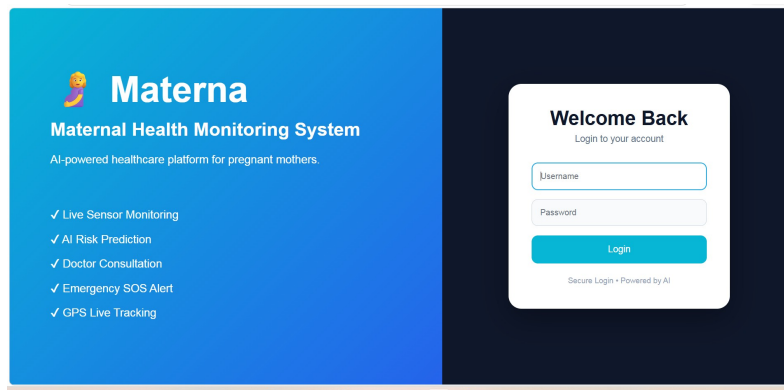


Figure 3.16: User login interface of the Materna AI system

3.5.2 Admin Dashboard

The admin dashboard provides an overview of the system. It also allows the administrator to manage doctors, patients, predictions, consultations, etc. In addition, it provides quick access to their management.

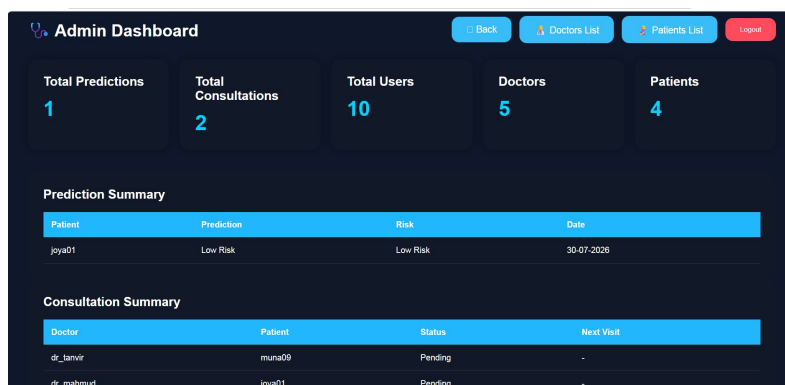
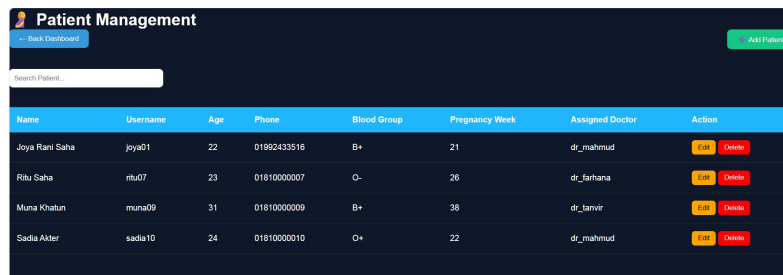


Figure 3.17: Admin dashboard of the Materna AI system

Name	Username	Phone	Specialization	Hospital	Experience	Qualification	BMDC	Action
Dr. Mahmud Hasan	dr_mahmud	01911111111	Gynecology	Dhaka Medical College Hospital	10 Years	MBBS, FCPS	A12345	Edit Delete
Dr. Tanvir Ahmed	dr_tanvir	01911111112	Obstetrics	Square Hospital	8 Years	MBBS, FCPS	A12346	Edit Delete
Dr. Farhana Islam	dr_farhana	01911111113	Gynecology	United Hospital	12 Years	MBBS, FCPS	A12347	Edit Delete
Dr. Nusrat Jahan	dr_nusrat	01911111114	Obstetrics	Evercare Hospital	9 Years	MBBS, FCPS	A12348	Edit Delete
Dr. Shameen Akter	dr_shameen	01911111115	Gynecology	Popular Hospital	15 Years	MBBS, FCPS	A12349	Edit Delete

Figure 3.18: Doctor management interface



The screenshot shows a 'Patient Management' dashboard. At the top, there's a 'Back Dashboard' button and an 'Add Patient' button. Below is a search bar labeled 'Search Patient'. The main part of the interface is a table with columns: Name, Username, Age, Phone, Blood Group, Pregnancy Week, Assigned Doctor, and Action. The table contains four rows of patient data. Each row has 'Edit' and 'Delete' buttons in the Action column.

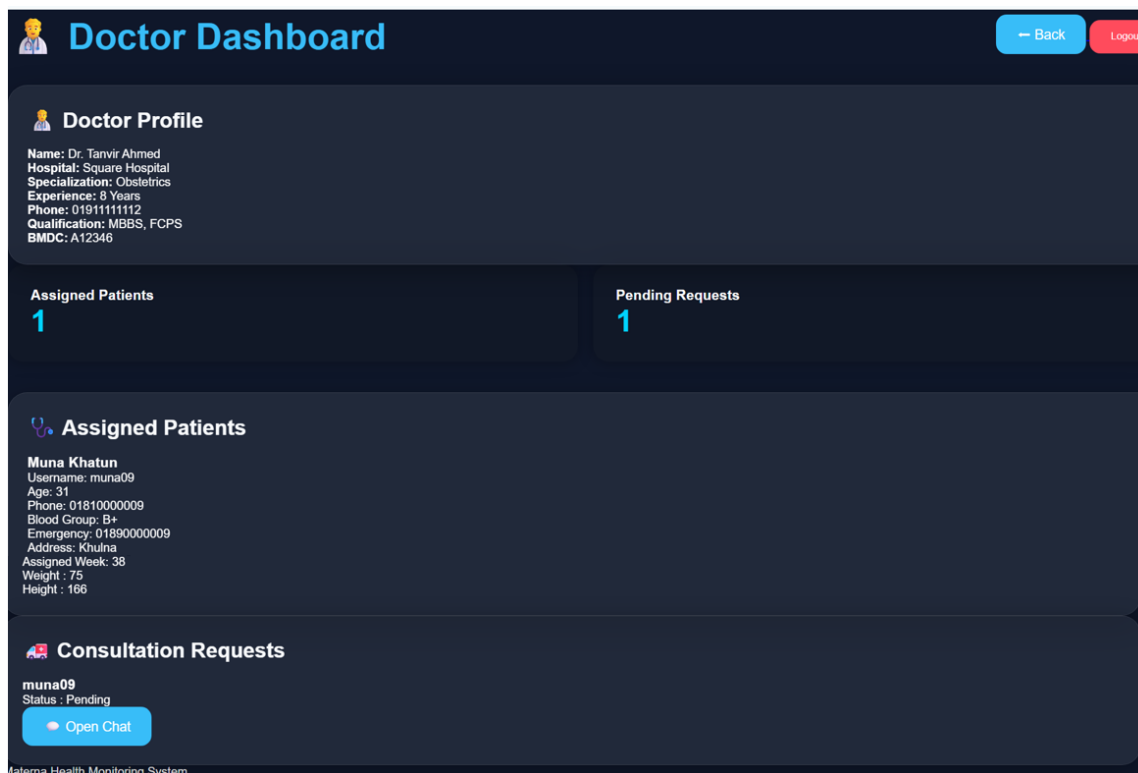
Name	Username	Age	Phone	Blood Group	Pregnancy Week	Assigned Doctor	Action
Joya Rani Saha	joya01	22	01992433518	B+	21	dr_mahmud	Edit Delete
Ritu Saha	ritu07	23	01810000007	O-	26	dr_fahana	Edit Delete
Muna Khatun	muna09	31	01810000009	B+	38	dr_tanvir	Edit Delete
Sadia Akter	sadia10	24	01810000010	O+	22	dr_mahmud	Edit Delete

Figure 3.19: Patient management interface

The doctor management interface enables the administrator to view, search, add, edit and delete doctors. In a like manner, the patient management interface provides access to the patient's age, blood group, week of pregnancy and doctor.

3.5.3 Doctor Dashboard

Doctors can have a look at their assigned patient's information and handle consultation inquiries. It also helps with communication with people through the consultation and messaging features.



The screenshot shows a 'Doctor Dashboard' for Dr. Tanvir Ahmed. At the top, there's a 'Back' button and a 'Logout' button. The dashboard is divided into several sections:

- Doctor Profile:**
 - Name: Dr. Tanvir Ahmed
 - Hospital: Square Hospital
 - Specialization: Obstetrics
 - Experience: 8 Years
 - Phone: 01911111112
 - Qualification: MBBS, FCPS
 - BMDC: A12346
- Assigned Patients:** 1
- Pending Requests:** 1
- Assigned Patients (Details):**
 - Muna Khatun
 - Username: muna09
 - Age: 31
 - Phone: 01810000009
 - Blood Group: B+
 - Emergency: 01890000009
 - Address: Khulna
 - Assigned Week: 38
 - Weight: 75
 - Height: 166
- Consultation Requests:**
 - muna09
 - Status: Pending
 - [Open Chat](#)

At the bottom, it says 'Materna Health Monitoring System'.

Figure 3.20: Doctor dashboard of the Materna AI system

3.5.4 Patient Dashboard

Patient dashboard in health parameters such, as blood pressure, temperature, heart rate and blood sugar so that the dashboard can perform risk prediction. Dashboard also shows sensor readings ECG analysis, fall detection status, GPS information and assigned doctor details. I find that dashboard is very useful. Emergency SOS button can be pressed to send an alert to the registered emergency contact.

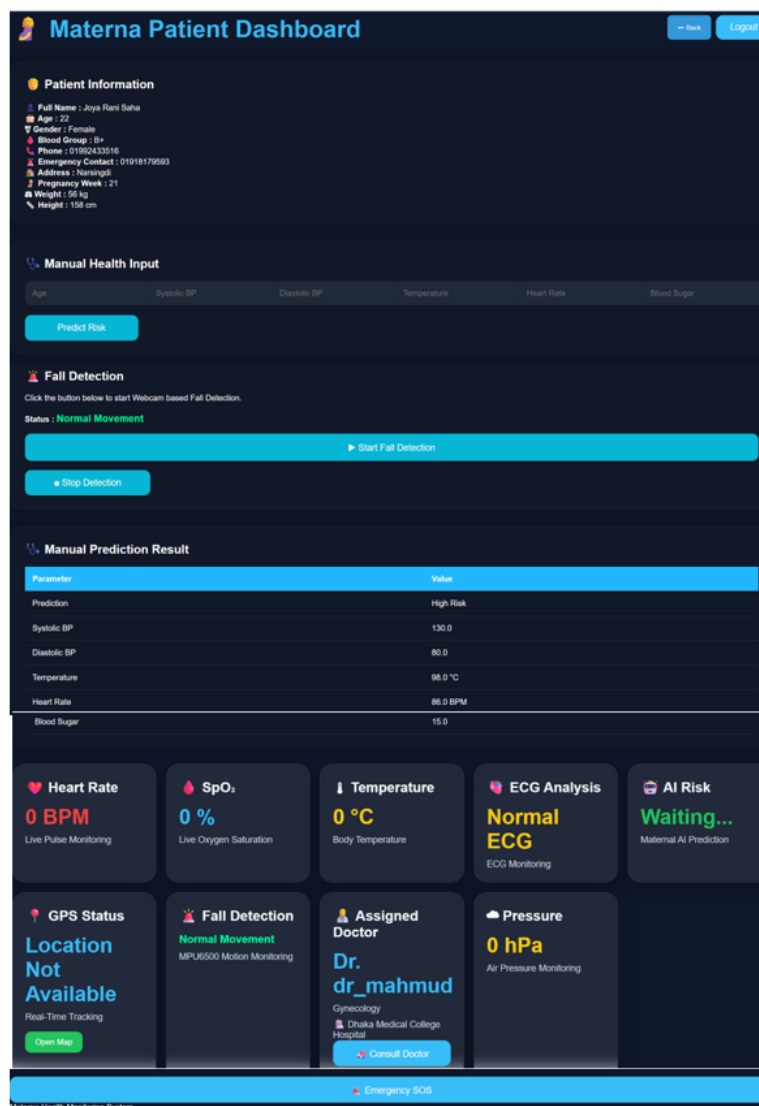


Figure 3.21: Patient dashboard showing real-time health monitoring and healthcare services

3.5.5 Fall Detection

An YOLO-based computer vision model was added to the Materna AI system to detect falls in time. The Materna AI system uses the laptop webcam to watch movement and gives a web-based interface that lets users start and stop fall detection. The system status shows the detected condition, for example normal movement or a fall event. Figure 3.22 shows the webcam-based fall detection interface of the proposed system.

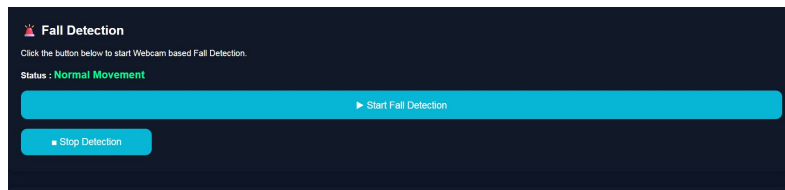


Figure 3.22: Webcam-based fall detection interface of the Materna AI system

3.5.6 Doctor Consultation System

The patient can click the consult doctor button to request a consultation, with the assigned doctor. The doctor can view the request. Communicate with the patient through the consultation system.



Figure 3.23: Doctor consultation interface in the Materna Health Monitoring System

3.5.7 GPS Tracking System

The GPS tracking system patients see their real-time location, on the web application. The GPS tracking system shows the GPS status. Gives an open map button so patients can view their location.

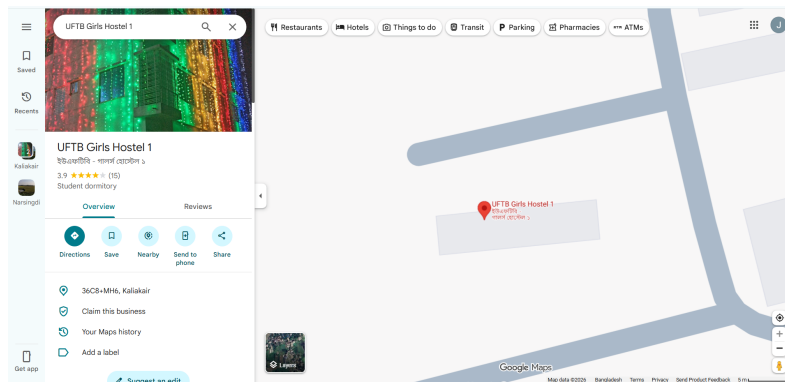


Figure 3.24: GPS status and location tracking interface

3.5.8 Emergency Alert and Notification System

The Materna AI system offers a way to send alerts when there is a serious problem with a mothers health. If a dangerous situation or a fall happens the system creates a message with the patients details and information, about the pregnancy. This message is sent to the doctors phone number so the doctor can see it is an emergency and do what needs to be done.

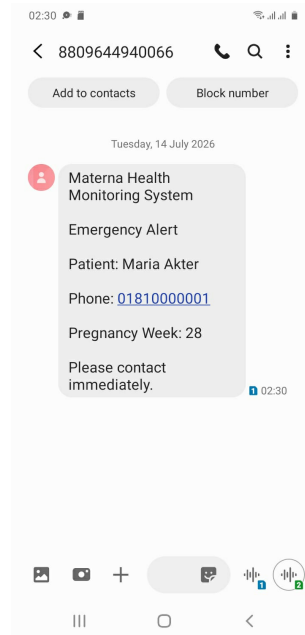


Figure 3.25: Materna AI web dashboard showing real-time maternal health monitoring and emergency SOS interface

Chapter 4

Machine Learning Model Development

4.1 Machine Learning Pipeline

4.1.1 Introduction

Machine learning plays a significant role in the proposed Materna system. The system proposed in this work is designed to make three health risk classifications for the mother: Low Risk, Mid Risk, and High Risk, rather than relying solely on manual assessment. Supervised machine learning algorithms are used to train models. The trained models arrive at a maternal health risk level based on important health factors such as maternal age, maternal blood pressure (systolic and diastolic), blood sugar level, maternal temperature, and maternal heart rate. Different supervised machine learning algorithms were trained and tested to determine the best model that can be used to predict maternal danger. Logistic Regression, Decision Tree, KNN, Gradient Boosting, XGBoost and Random Forest are the models evaluated. They were tested by using the standard classification metrics and the best model was chosen to be integrated into the proposed system. Optimization and deployment of the selected Random Forest model was then done using a web application with Flask for real-time maternal health risk prediction. This machine learning pipeline is used to make predictions about maternal health, which is part of the proposed Materna system.

4.1.2 Dataset Description

The proposed maternal health risk prediction model was developed using the Maternal Health Risk dataset. The data set is composed of physiological values measured for pregnant women and are intended to be used for classification of maternal health statuses into three categories: Low Risk, Mid Risk and High Risk. The input features selected are: maternal age, systolic blood pressure, diastolic blood pressure, blood sugar level, body temperature and heart rate. These indicators have been chosen because they are considered to be important parameters to evaluate maternal health status.

Table 4.1: Features of the Maternal Health Risk Dataset

Feature	Data Type	Description
Age	Integer	Age of the pregnant woman (years).
SystolicBP	Integer	Systolic blood pressure (mmHg).
DiastolicBP	Integer	Diastolic blood pressure (mmHg).
BS	Float	Blood sugar level.
BodyTemp	Float	Body temperature measurement.
HeartRate	Integer	Heart rate (beats per minute).
RiskLevel	Categorical	Maternal risk class (Low, Mid, High).

The data set has six physiological input features and a one target variable RiskLevel. The outcome is a classification of each maternal record as Low Risk, Mid Risk or High Risk. The dataset was used to train and evaluate the supervised machine learning models. The maternal health records were split into an 80:20 training and testing set for the development of the model. These machine learning models were trained on the training subset and tested on the testing subset in order to assess the models' prediction capabilities. Table 4.2 shows the distribution of the data set.

Table 4.2: Distribution of the Maternal Health Dataset

Dataset Split	Number of Samples	Percentage
Total Dataset	5,490	100%
Training Set	4,392	80%
Testing Set	1,098	20%

In total, the dataset comprises 5,490 samples, of which 4,392 samples are used for the training set, and 1,098 samples for the test set. This 80:20 split is adequate to provide enough data for model training and also leave an independent set for model performance

evaluation.

In addition, the class distribution of the maternal health risk dataset was examined to gain an understanding of the distribution of the three categories of health risk. Table 4.3 displays the number of samples in each class for this data set: 2,215 samples are classified as Low Risk, 1,766 as Mid Risk, and 1,505 as High Risk. The High Risk class is the smallest and the Low Risk class is the largest proportion of the data set.

Table 4.3: Class Distribution of the Maternal Health Risk Dataset

Risk Level	Number of Samples	Percentage
Low Risk	2,215	40.38%
Mid Risk	1,766	32.19%
High Risk	1,505	27.43%
Total	5,490	100%

The class distribution can give an overview of the representation of maternal health risk levels in the data. The supervised machine learning models were trained and evaluated using the dataset, with the testing set being reserved for independent evaluation.

4.1.3 Data Preprocessing

The maternal health data was preprocessed to guarantee the reliability of prediction performance of the machine learning models before training the models. The data was first loaded into Python within the Pandas library and checked for any missing or conflicting data. The data set chosen was clean, with no large amount of missing data, therefore no imputation technique was needed.

The dataset comprised of six features, namely Age, Systolic Blood Pressure, Diastolic Blood Pressure, Blood Sugar, Body Temperature, and Heart Rate. The target variable, RiskLevel was converted to numerical labels to be used with supervised machine learning algorithms.

Lastly, a train-test split was applied to the data set to prepare the training and testing

sets. Machine learning models were developed using the training data, and tested on the testing data to assess how well they predict.

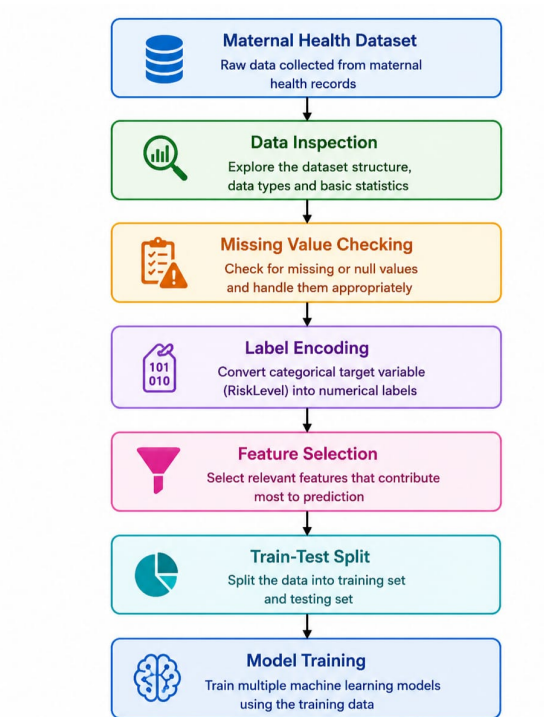


Figure 4.1: Data preprocessing workflow used for maternal health risk prediction.

4.1.4 Machine Learning Algorithms

Table 4.4: Performance Comparison of Machine Learning Models

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.6384	0.62	0.63	0.61
Decision Tree	0.9299	0.93	0.94	0.93
K-Nearest Neighbors	0.9089	0.91	0.91	0.91
Gradient Boosting	0.8770	0.88	0.88	0.88
XGBoost	0.9162	0.92	0.92	0.92
Random Forest	0.9308	0.93	0.94	0.93

The highest accuracy evaluated machine learning model was Random Forest with an accuracy of 93.08%, precision of 93%, recall of 94%, and F1 score of 93%. Random Forest was chosen as the final model for Materna system due to its overall performance in terms of prediction and its suitability for real time deployment. It is also ensemble based, which means it also generalizes better than a single decision tree and does not suffer from overfitting as much.

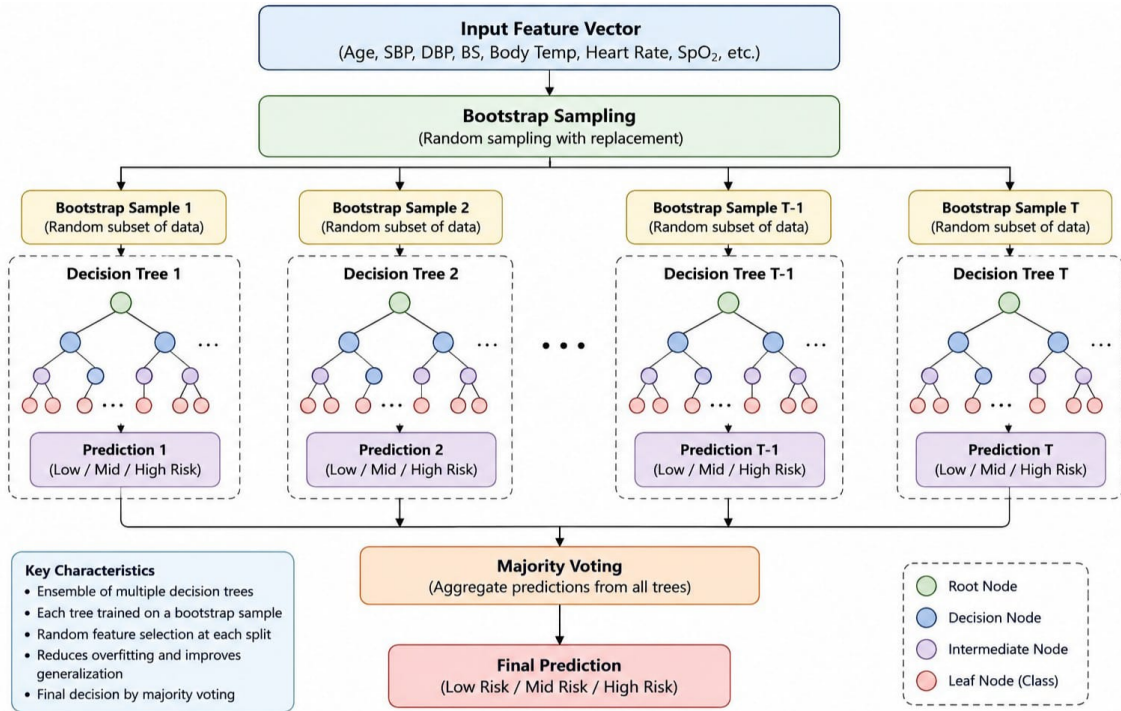


Figure 4.2: Architecture of the Random Forest classifier used for maternal health risk prediction.

The working principle of the Random Forest classifier applied by the proposed Materna system is depicted in figure 4.2. The algorithm starts by creating several bootstrap samples of the original training data set. A random subset of input features are selected at each split, and a separate decision tree is trained on each bootstrap sample. During inference, each decision tree makes a separate prediction of the maternal health risk category. Finally the forecasts of all trees are aggregated together to give the final risk classification by using majority voting. It is an ensemble learning method that extends the accuracy of prediction, decreases overfitting and increases the generalization ability of a single decision

tree.

Random Forest is a supervised ensemble learning algorithm, which is an integration of various decision trees to decrease overfitting and increase classification accuracy. Each decision tree is learned with a random subset of the training set, and predictions are made by combining the predictions of all the trees. The prediction function of the Random Forest classifier is given as:

$$f(x) = \frac{1}{T} \sum_{t=1}^T h_t(x) \quad (4.1)$$

Here T is the number of trees in the forest and $h_t(x)$ is the prediction from the t -th decision tree.

In the classification tasks, the final output is obtained by majority voting of all decision trees. The class to be predicted is calculated as :

$$\hat{y} = \text{mode} \{h_1(x), h_2(x), \dots, h_T(x)\} \quad (4.2)$$

where $\hat{y}$ is the final predicted maternal health risk class (Low Risk, Mid Risk, or High Risk). This strategy of ensemble helps to get more stable predictions and generalisation than a single decision tree.

4.1.5 Model Selection

Several supervised machine learning algorithms was trained and assessed for appropriateness of the model for maternal health risk prediction. Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), Gradient Boosting, XGBoost, and Random Forest were evaluated algorithms. The accuracy, precision, recall and F1-score were used as the metrics for evaluating the performance of each model.

Among the evaluated models, the Random Forest model (93.08%) had the best prediction accuracy along with a precision of 93%, recall of 94%, and F1-score of 93% . The Random Forest classifier was chosen as the final prediction model for deployment in the proposed Materna system based on its overall performance in prediction. Random Forest is robust, low complexity, fast, easy to maintain and supports better real-time health-

care applications while achieving high prediction performance. Its features are ideal for integration with an ESP32-powered IoT monitoring system and the Flask-based web application. Further efforts were made to improve the prediction performance by tuning the hyperparameters of the model with GridSearchCV by using five-fold cross-validation. The hyperparameters of the Random Forest model that were found to work best are given in Table 4.5. The optimized model was then saved as a serialised `maternal_model.pkl` file and embedded into the proposed maternal health risk prediction system in real-time.

Table 4.5: Optimized Hyperparameters of the Random Forest Model

Parameter	Value
Number of Trees (n_estimators)	300
Maximum Depth (max_depth)	15
Maximum Features (max_features)	sqrt
Minimum Samples Split	10
Minimum Samples Leaf	1

4.1.6 Model Deployment

The best and optimized Random Forest classifier was then serialized using the Pickle library in python and saved as `maternal_model.pkl`. The saved model is loaded by the Flask backend on system start.

The Flask server preprocesses input data in a similar manner to that preprocess during model training, when the user submits maternal health data via the web application. The processed feature vector is then fed into the trained Random Forest model to forecast the level of maternal health risk. The model categorizes the maternal health record into one of three categories: Low Risk, Mid Risk, or High Risk.

Finally, the result of the prediction is returned via REST API and presented on the web dashboard. This approach allows for the ability to predict maternal health risks in real-time and with high accuracy, and also ensures consistency in the training and inference process.

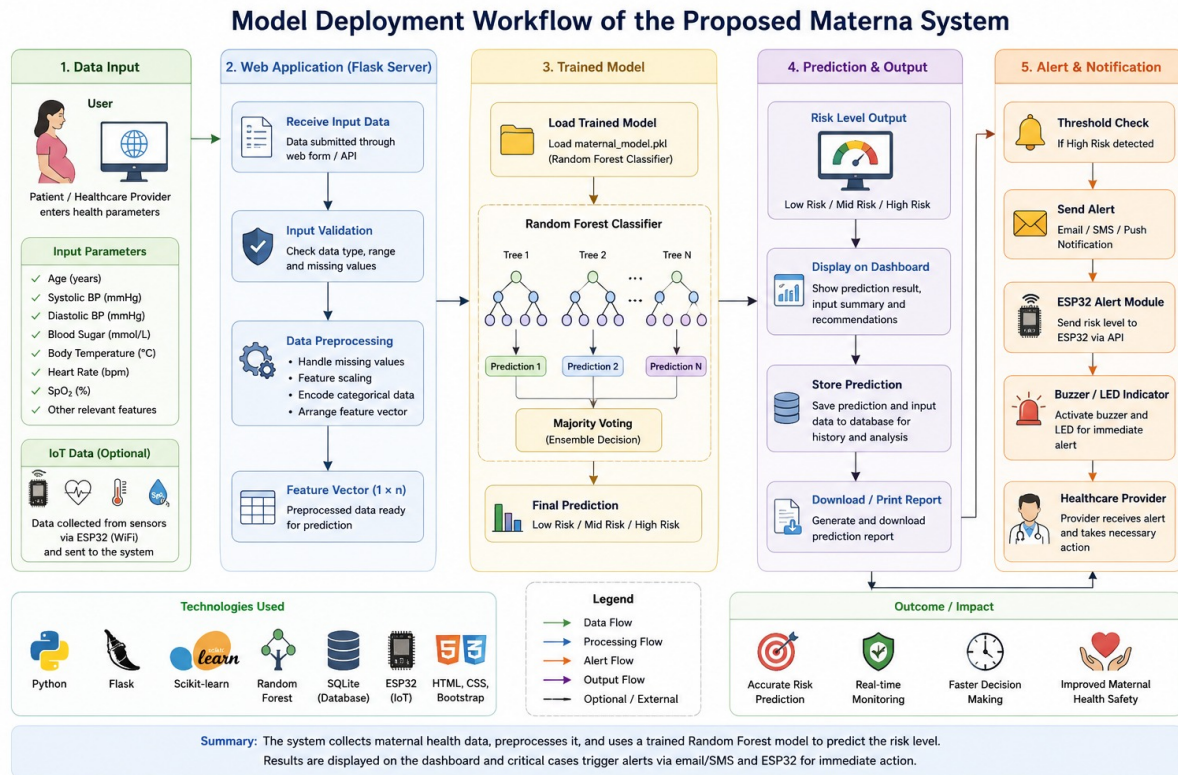


Figure 4.3: Model deployment workflow of the proposed Materna system.

The entire deployment procedure of the proposed Materna system is shown in Figure 4.3. The Flask web application takes the parameter for maternal health from the user, validates and preprocesses the input, and forms a feature vector, which is then used to predict the user's health status. The trained Random Forest model (`maternal_model.pkl`) is then loaded to classify the maternal health risk as Low Risk, Mid Risk or High Risk. The prediction result is reported to the web dashboard and can be utilized by the monitoring module based on ESP32 for timely alerts of high-risk cases.

The deployed architecture integrates the trained Random Forest model with the Flask-based web application to provide efficient, scalable, and real-time maternal health risk prediction for the proposed Materna system.

The architecture implemented combines the trained Random Forest model with the web application developed using Flask framework, ensuring efficient, scalable, and real-time maternal health risk prediction for the proposed Materna system.

4.1.7 Performance Evaluation Metrics

Standard classification metrics were used to assess the proposed maternal health risk prediction model. The accuracy was used to measure the overall correctness of the classifier while precision, recall, and F1-score were used to assess the quality of predictions for maternal health risk class. In addition, the Receiver Operating Characteristic (ROC) curve and Area Under the Curve (AUC) were used to further evaluate the classification performance of the Random Forest model. These are all good measures of how well the model predicts.

Accuracy is calculated as

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.3)$$

Precision is defined as

$$Precision = \frac{TP}{TP + FP} \quad (4.4)$$

Recall is defined as

$$Recall = \frac{TP}{TP + FN} \quad (4.5)$$

Finally, the F1-score is computed as

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4.6)$$

where TP , TN , FP , and FN represent True Positive, True Negative, False Positive, and False Negative, respectively.

In addition the Receiver Operating Characteristic (ROC) curve was utilized to assess the ROC model's performance in separating the maternal health risk groups. The ROC curve is a plot of the True Positive Rate against the False Positive Rate for various thresholds set for a classification. The Area Under the Curve (AUC) is a summary measure of the model's discriminatory performance with higher AUC values being better

class discriminators.

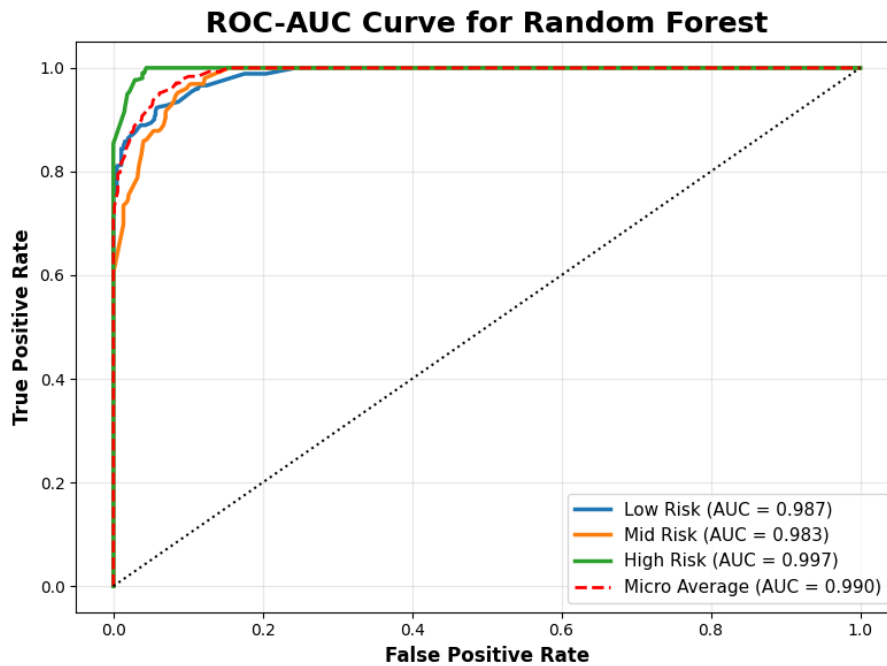


Figure 4.4: ROC-AUC curves of the Random Forest model for maternal health risk classification.

The ROC-AUC curves of the three maternal health risk classes using the Random Forest model are shown in Figure 4.4. The model was able to obtain AUC values of 0.987, 0.983 and 0.997 for the Low Risk, Mid Risk, and High Risk categories, respectively, while achieving an AUC micro-average of 0.990. The high AUC values suggest that the Random Forest model can efficiently classify the various maternal health risk groups.

Table 4.6: Classification Report of the Random Forest Model

Class	Precision	Recall	F1-score	Support
Low Risk	0.96	0.90	0.93	443
Mid Risk	0.87	0.93	0.90	349
High Risk	0.96	0.98	0.97	306
Macro Average	0.93	0.94	0.93	1098
Weighted Average	0.93	0.93	0.93	1098

Table 4.6 shows overall classification performance of the Random Forest model. The precision, recall and F1 score of the classifier were high in all maternal health risk categories. The highest recall value was 0.98 for the High Risk Class and the highest F1 score was 0.97, showing its potential to correctly identify records in the High Risk class.

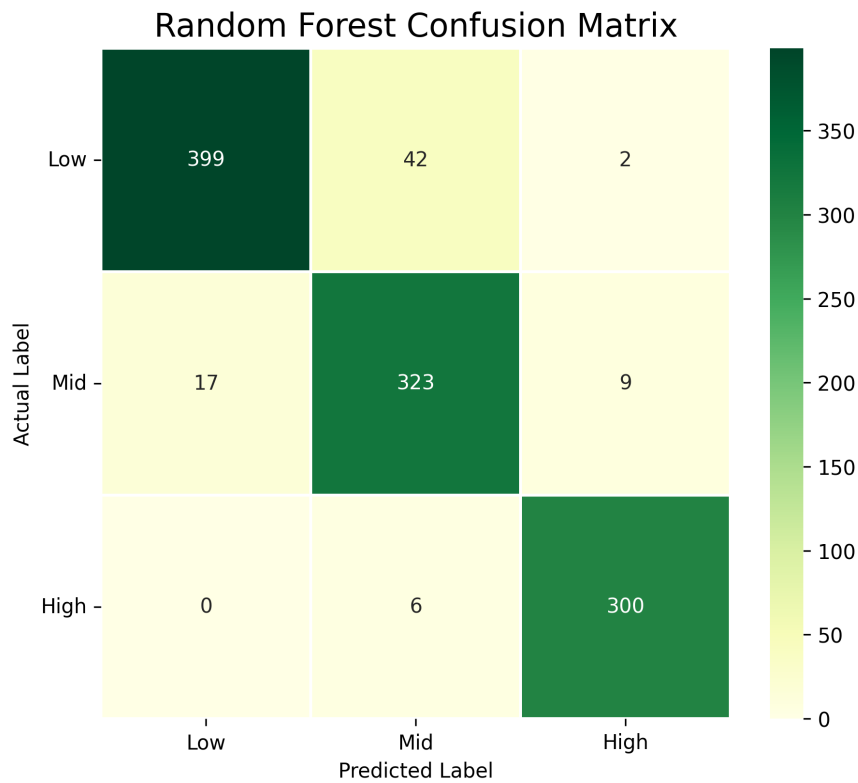


Figure 4.5: Confusion Matrix of the Random Forest classifier.

The confusion matrix of the Random Forest Classifier is presented in figure 4.5. The main diagonal of the matrix shows a significant proportion of samples predicted correctly, suggesting good prediction accuracy. There are 399 samples correctly classified for the Low Risk class, 323 for the Mid Risk class and 300 for the High Risk class. The misclassified samples between risk categories are few, especially High Risk class for good classification performance.

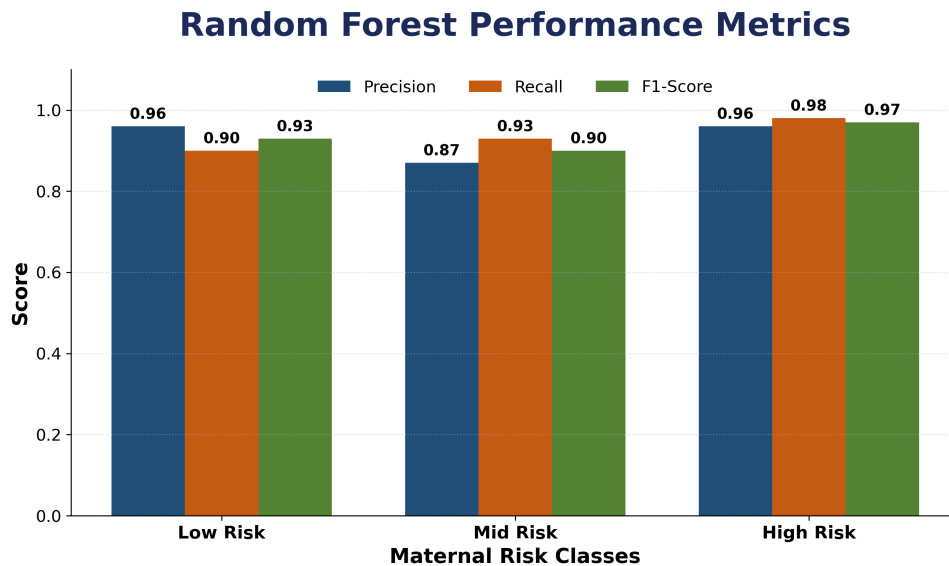


Figure 4.6: Precision, Recall, and F1-score of the Random Forest model for each maternal health risk class.

The precision, recall and F1-score of the Random Forest classifier are shown in the figure 4.6 for each maternal health risk category. The best performance was obtained in the High Risk class with a precision of 0.96, recall of 0.98 and F1-score of 0.97. The outcomes illustrate that the model has a consistent performance in the three maternal health risk categories.

The final Random Forest classification had an overall prediction accuracy of 93.08% on the maternal health dataset. This result shows the accuracy of the model used for classification of maternal health risk as Low Risk, Mid Risk, and High Risk.

4.2 Human Fall Detection Model Development

4.2.1 Introduction

The human fall-detection module was designed to detect the fall of a person from a real stream of camera images. This system continuously takes frames from a camera and applies a YOLO model of object detection to each frame. The detected person is then categorized into two target classes – *Fall* and *Non-Fall*. This real-time detection is designed to aid in identifying fall events in time.

6446 images of human falls were annotated, and used for both training and evaluation of the fall detection module. The dataset has two classes, Fall and Non-Fall and contains bounding-box annotations for object detection. Four variants of YOLO, namely YOLOv8n, YOLOv9t, YOLO11s and YOLO26n, were trained and tested with the same dataset and training conditions to find an appropriate model for real-time deployment.

The trained models were tested with common object detection metrics such as precision, recall, mAP@50, mAP@50-95 and inference time. Their performance was compared to find the most appropriate model for live camera based fall detection. YOLOv8n was chosen as the final deployment model for its' high recall rate of 99.7% and its' fast inference time of 2.3 ms per image according to the experiment results, which meets the requirements of image identification task and real-time application.

The chosen model YOLOv8n is used in the final implementation along with a live camera through OpenCV. The camera feeds video frames into the detection model which generates a prediction of whether it is a *Fall* or *Non-Fall* event, and outputs the predicted bounding box and confidence score shown on the screen as real time data.

4.2.2 Dataset Description

The proposed real time fall detection module was developed and tested with a human fall detection dataset. The dataset was obtained from Roboflow and contains a total of 6,446 annotated images. The data set has two possible target classes: *Fall* and *Non-Fall*. The images include the bounding-box information, which helps the YOLO-based object detection models determine the class and position of the detected person.

The data set was split into training, validation and testing sets. The total number of images for training were 3,435, validation were 859, and testing were 2,152 images. The proportions of the training set, validation set and testing set were 53.29%, 13.33%, and 33.39%, respectively. To build and test the models, all images were resized to 640×640 pixels.

The *Fall* class has 2,256 annotations and the *Non-Fall* class has 4,190 annotations. Thus, the proportion of annotations in the two classes (Fall and Non-Fall) is 35.00% and 65.00%,

respectively. Distributions of the detailed data set are shown in Table 4.7.

Table 4.7: Dataset Distribution for Human Fall Detection

Dataset Split	Number of Images	Percentage
Training	3,435	53.29%
Validation	859	13.33%
Testing	2,152	33.39%
Total	6,446	100%

Table 4.8: Class Distribution of the Human Fall Detection Dataset

Class	Annotations	Percentage
Fall	2,256	35.00%
Non-Fall	4,190	65.00%
Total	6,446	100%

As shown in the data distribution, most of the images are from the class *Non-Fall* with a smaller but significant number of images from *Fall* class. Although the two classes differ in class distribution, there is still enough ground-truth annotated samples to train and test the object detection models.

The training, validation and testing sets were kept separate in the model building process, and the consistency of the evaluation of trained models was ensured. For last performance evaluation the test set was reserved.

4.2.3 Dataset Preprocessing

The data was preprocessed for the object detection task before training the fall detection models based on YOLO. The images and the bounding-box annotations were arranged in the YOLO dataset format. A person was annotated with a class label and the normalized bounding-box coordinates of the detected person in an annotation file for each image.

There were two classes of object detection in the dataset: *Fall* and *Non-Fall*. The class labels were translated into numerical labels for training with YOLO: class 0 was converted

to *Fall* and class 1 was converted to *Non-Fall*. To keep the input size the same for all the YOLO models, the images were resized to 640×640 pixels.

The prepared dataset was partitioned into separate training set, validation set and testing set. The training images and the related annotations were utilized for learning the model, and the validation subset was for tracking model performance during training. The test set was not used in model training, but was reserved only for the final evaluation of the trained models.

The path to the datasets, number of classes and class names were defined via a dataset configuration file that was required by the YOLO training framework. This resulted in a structure of the data that allowed to train and compare the same input format and data distribution for YOLOv8n, YOLOv9t, YOLO11s and YOLO26n.

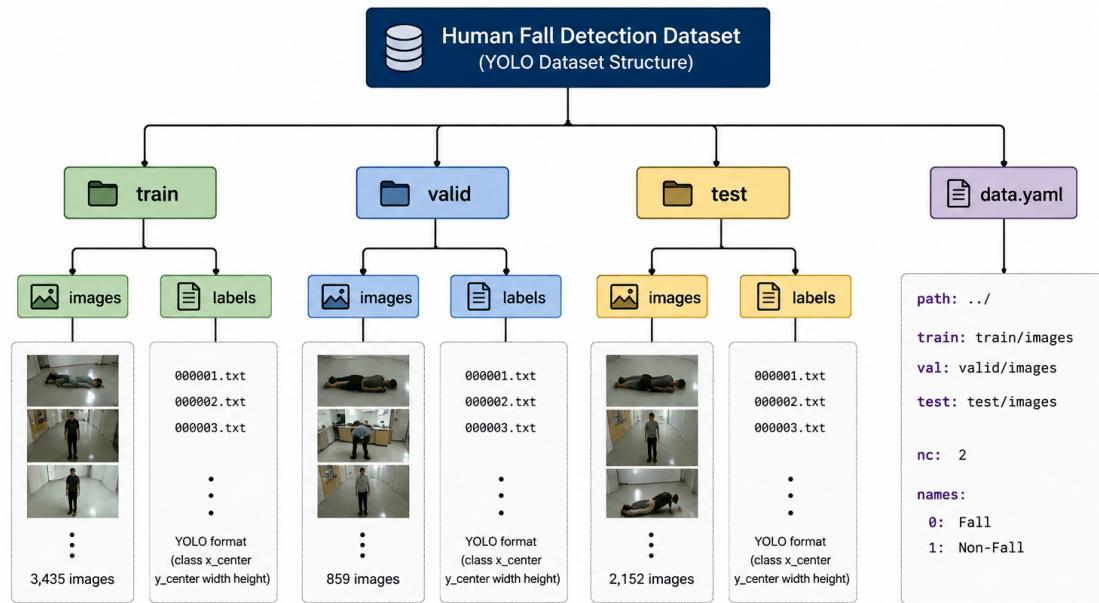


Figure 4.7: YOLO-based dataset structure used for human fall detection.

The distribution of object annotations in the training set was also analyzed to understand the class balance and bounding-box characteristics. Figure 4.8 presents the number of annotated instances for the *Fall* and *Non-Fall* classes, along with the spatial distribution and size of the bounding boxes in the training images.

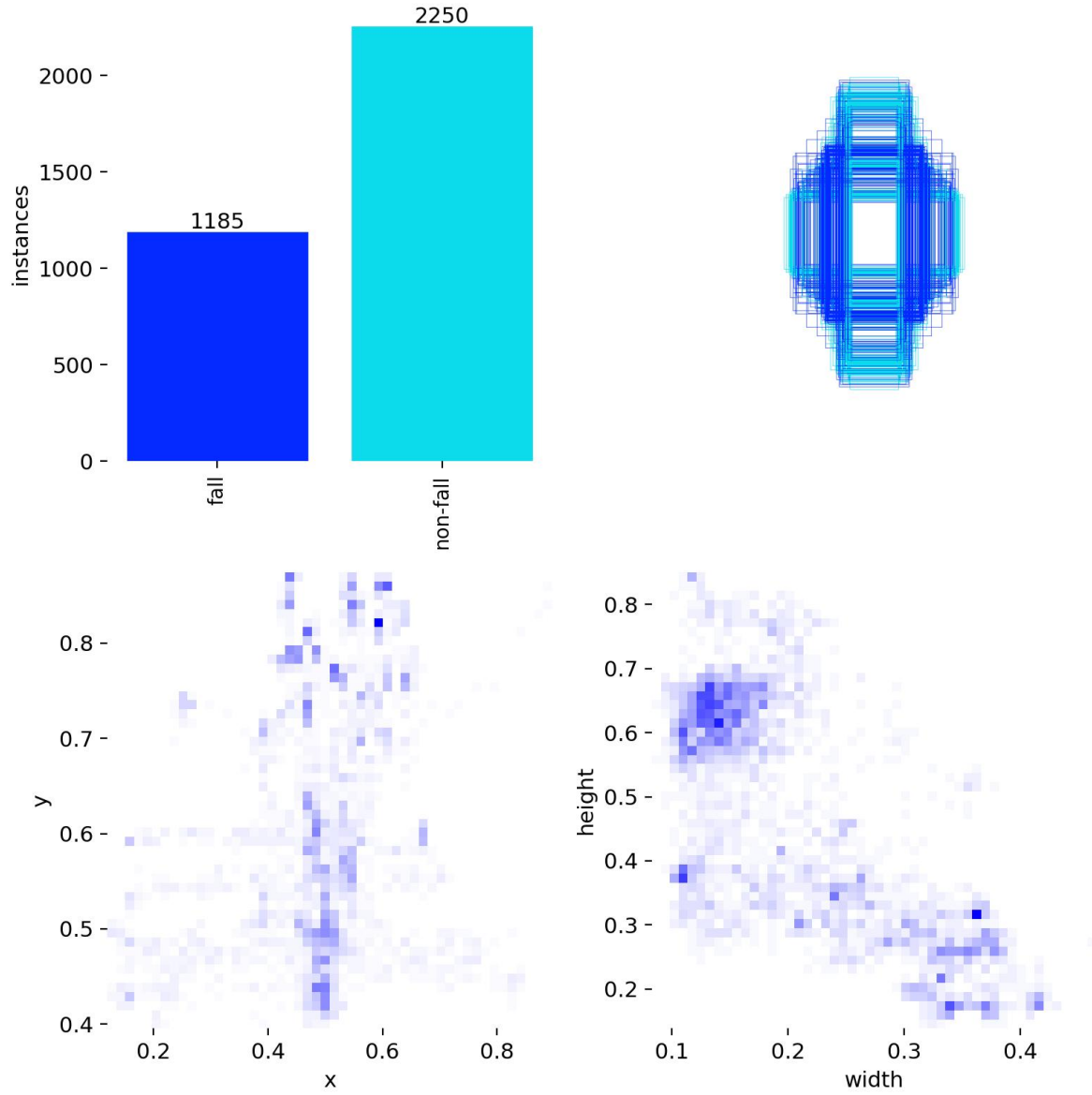


Figure 4.8: Class and bounding-box distribution of the training set.

The training set contains 1,185 *Fall* instances and 2,250 *Non-Fall* instances. The visualization also shows the spatial distribution of the annotated bounding boxes and their relative widths and heights. This analysis helps verify the annotation quality and provides an overview of the object locations and sizes used during YOLO model training.

4.2.4 YOLO Model Training

The prepared human fall detection dataset was used to train four YOLO based object detection models: YOLOv8n, YOLOv9t, YOLO11s and YOLO26n. To make the results

of the models comparable, they were trained using the same set of data and the same image processing environment. An image resolution of 640×640 pixels was used for processing the input images.

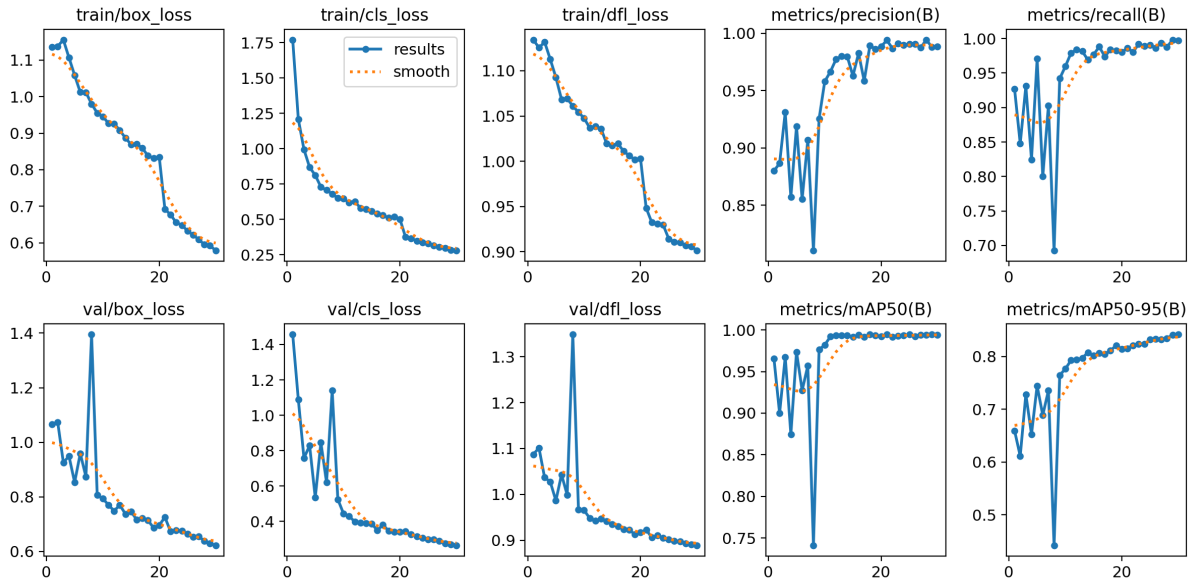


Figure 4.9: Training and validation results of the YOLOv8n fall detection model.

The annotated bounding boxes were used to train the models to classify two target classes (Fall and Non-Fall). Training loss and validation loss were used to monitor the training process with other detection performance metrics like precision, recall, mAP@50, and mAP@50-95.

For YOLOv8n model, the curve for training and the metrics for detection obtained during training are displayed in Figure 4.9. Figure illustrates how training and validation loss, and precision, recall, mAP@50 and mAP@50-95 change with the training epochs.

4.2.5 Model Performance Evaluation

Standard object detection metrics (precision, recall, mAP at 50, and mAP at 50-95) were used for the performance evaluation of the trained YOLO models. The models were tested on how well they can detect and classify the classes *Fall* and *Non-Fall* using these metrics. Precision is the percentage of the objects correctly identified by the model, and recall is the percentage of objects accurately identified by the model. mAP@50 is based on

the Intersection over Union (IoU) threshold of 0.50 and mAP@50-95 is a more holistic assessment that calculates the mean average precision at various IoU thresholds ranging from 0.50 to 0.95.

The experimental results showed the model yolov8n select in this article had excellent detection performance. The model was able to identify almost all the instances of true fall and non-fall with a recall of 99.7%, meaning that the model is able to capture nearly all the true fall and non-fall cases. In addition, the model exhibited excellent accuracy and mAP results, indicating its ability to accurately locate and classify objects. The mAP@50 and mAP@50-95 values derived from the training and evaluation are reported in the performance results.

The confusion matrix was also analyzed to look into the classification behavior of the YOLOv8n model in the target classes. It represents in detail the correct and incorrect predictions of the classes *Fall* and *Non-Fall*.

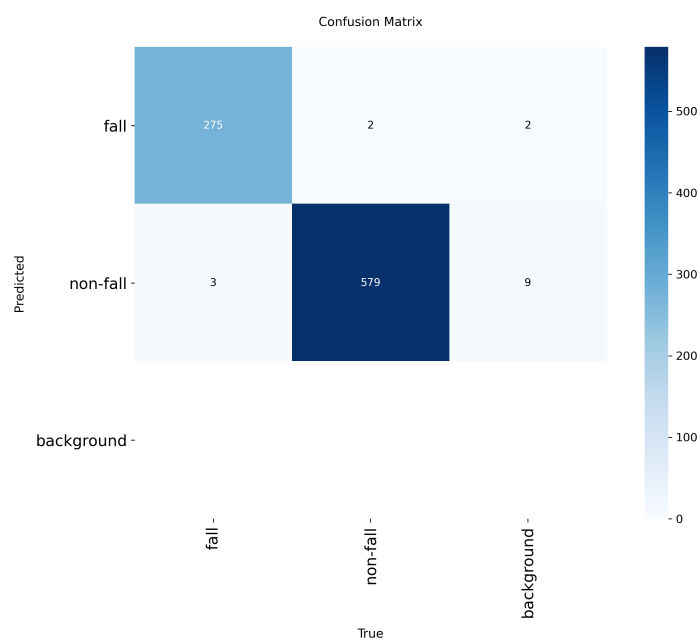


Figure 4.10: Confusion matrix of the YOLOv8n fall detection model.

The confusion matrix of the YOLOv8n model on the *Fall* and *Non-Fall* classes is shown in figure 4.10. The matrix shows the number of correctly and incorrectly detected instances for each class. The more correct predictions along the diagonal the better the model is

able to differentiate the fall events from the non-fall events.

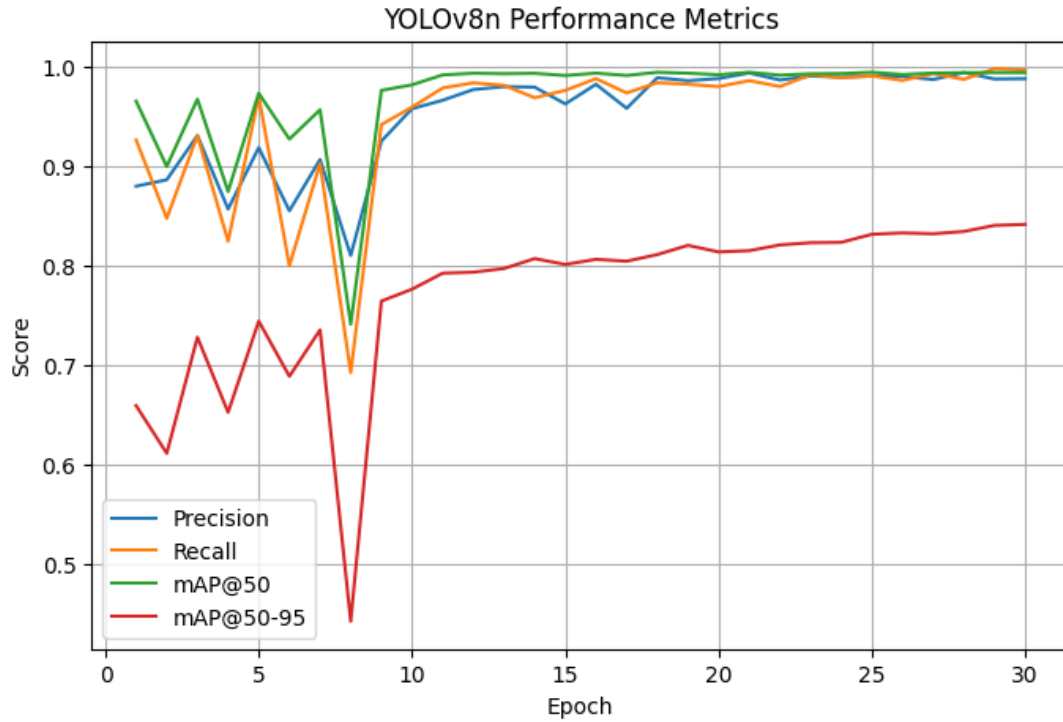


Figure 4.11: Training performance metrics of the YOLOv8n model.

The results demonstrate that the model performance increases significantly when it is trained. Precision and recall increase and stabilize at high values in the later epochs. Also, the mAP@50 is close to 1.0, which means that the object detection performance is good at IoU 0.50. The mAP@50–95 is slowly improved during training, achieving roughly 0.84 at epoch 33, indicating satisfactory performance in localization tasks with more stringent IoU requirement.

The relatively good performance from the last few epochs suggests the model has successfully identified the distinguishing features of the Fall and Non-Fall classes. The results confirm that YOLOv8n is capable of real-time fall detection with a live camera input.

4.2.6 Comparative Analysis of YOLO Models

Four models based on YOLO were trained and tested: YOLOv8n, YOLOv9t, YOLO11s, and YOLO26n, all with the same experimental conditions and human fall detection dataset. To compare the models, the precision, recall, and mAP@50 and mAP@50-95

were used to determine the detection performance of the models.

Table 4.9 summarizes the results of the YOLO models tested. It was seen that all four models had high precision and recall values which shows that they are able to detect the class of *Fall* and *Non-Fall* effectively. The mAP@50 values were also near to 1.0 for all models indicating good object detection results at IoU threshold 0.50.

Table 4.9: Performance Comparison of the Evaluated YOLO Models

Model	Precision	Recall	mAP@50	mAP@50–95
YOLOv8n	98.9%	99.7%	99.4%	84.1%
YOLOv9t	99.6%	99.2%	99.4%	84.7%
YOLO11s	98.9%	99.2%	99.4%	84.1%
YOLO26n	98.1%	98.8%	99.4%	84.4%

Figure 4.12 provides a graphical comparison of the precision, recall, mAP@50, and mAP@50–95 values of the four models.

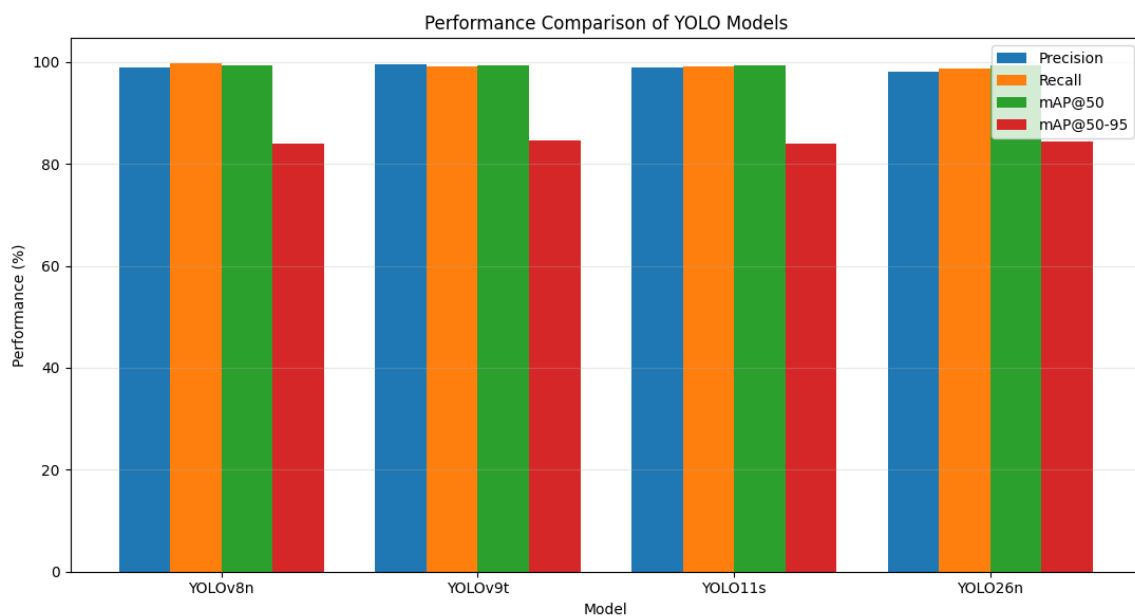


Figure 4.12: Performance comparison of the evaluated YOLO models.

YOLOv9t had the best result in terms of precision with a result of 99.6% and also had the best performance in the mAP@50–95 with a result of 84.7%. Among the tested models,

however, YOLOv8n had the highest recall (99.7%), suggesting it identified the largest percentage of instances of falls and non-falls from the total population of falls in the population. The recall of YOLO11s was also 99.2%, and for YOLO26n it was 98.8%.

Recall is a key measure for a safety-based fall detection system as a false negative can lead the system to miss a true fall event. The models evaluated were, therefore, further compared for their recall performance as illustrated in: Figure 4.13.

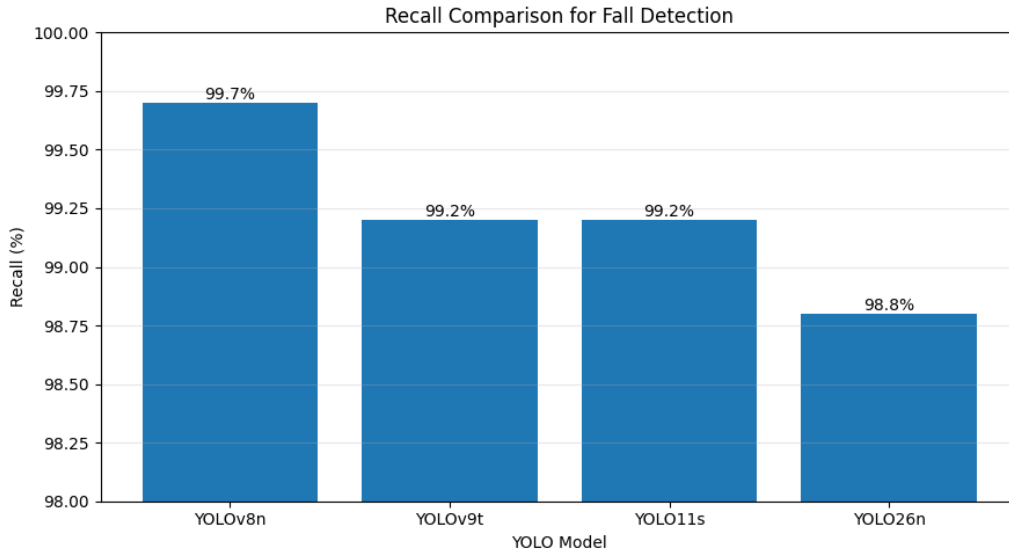


Figure 4.13: Recall comparison of the evaluated YOLO models for fall detection.

Overall, all four YOLO models showed good performance in detecting human falls from the human fall detection dataset. While the precision and mAP@50–95 of YOLOv9t were slightly higher, the highest recall was obtained by YOLOv8n among the tested models. The final model was therefore chosen, considering not only the detection performance, but also the computational efficiency, which is discussed further below.

4.2.7 Final Model Selection

After comparing all the models, YOLOv8n was chosen as the final model for the real-time fall detection module of the proposed Materna system. Although the evaluated models, YOLOv8n obtained the highest recall of 99.7% but YOLOv9t got slightly better precision and mAP@50–95. However, because fall detection is a safety application, it is important that the high percentage of true fall events is not missed, with recall being important.

Besides the detection performance, inference speed was also found to be an important factor that is crucial when it comes to real-time deployment. The inference time of the four evaluated YOLO models is shown in figure 4.14.

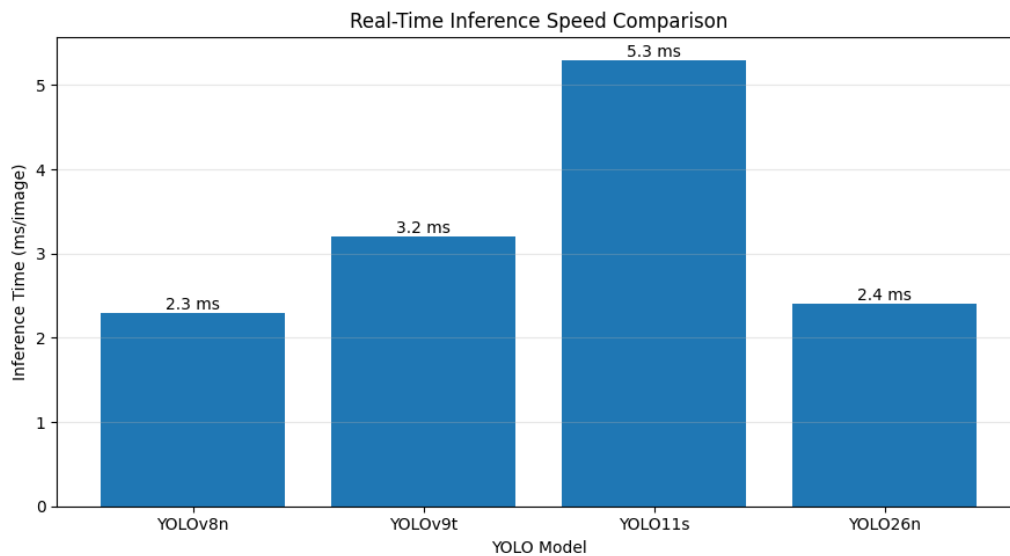


Figure 4.14: Inference speed comparison of the evaluated YOLO models.

YOLOv8n had the lowest inference time of 2.3 ms/frame as shown in Figure 4.14. YOLO26n needed 2.4 ms and YOLOv9t and YOLO11s needed 3.2 ms and 5.3 ms, respectively. YOLOv8n has a low inference time, which is suitable for processing live camera frames with a minimum latency.

After evaluating the models, YOLOv8n was chosen as the final model for the proposed real-time fall detection system, as it had the highest recall, performed best overall, and required the least inference time. The chosen model was then applied to a video stream and OpenCV to detect *Fall* and *Non-Fall* events in real-time.

4.2.8 Real-Time Fall Detection

YOLOv8n was chosen as the final detection model and then it was connected to a live camera with the help of the OpenCV library for real-time fall detection. The camera continuously takes photos to create video frames, and the trained YOLOv8n model is used to detect objects in each frame.

The model is able to recognize the person detected in each frame and predict one of

the two target classes: *Fall* or *Non-Fall*. The result is shown by drawing a rectangular box around the person detected and by showing its predicted class label and confidence percentage. This enables the system to give an instant visual feedback of the detected condition.

The real-time detection process involves the continuous acquisition of frames, inference of the YOLOv8n model, object localization, class prediction, and result visualization. The chosen model YOLOv8n is capable of achieving a high recall rate and low inference time, making it ideal for real-time performance on live camera feeds with minimal latency.

4.2.9 Model Deployment

Selected YOLOv8n model was trained, evaluated, and tested in real-time, and was ready for deployment in the proposed Materna system. The trained model weights were stored in the `best.pt` file and loaded from the file using the Ultralytics YOLO framework.

The trained model (YOLOv8n) was then combined with the camera interface, which was built using OpenCV, for real-time operations. The camera takes frames of video at all times, and these frames are sent to the trained model for inference. The model detects the person and classifies the detected condition as either *Fall* or *Non-Fall* with the corresponding bounding box and confidence score.

The deployed model runs on the system, eliminating the need for a separate cloud-based inference service to execute the fall detection process. This reduces the delay in communication and allows for real time monitoring using the live camera feed.

The input image size and class configuration is the same as when the model is trained. Thus, the trained model can be directly used for inference on live camera frames without any inconsistencies between the training and inference stages.

Finally, YOLOv8n was chosen for deployment since it had a high recall of 99.7% and a low inference time of 2.3 ms per image. The model is designed to be used in real-time human fall detection in the proposed Materna system, which is essential for the system's functionality.

4.3 Chapter Summary

In this chapter, the development and evaluation of two AI based modules of the proposed Materna system were presented. Supervised machine learning algorithms such as Logistic Regression, Decision Tree, K-Nearest Neighbors, Gradient Boosting, XGBoost and Random Forest were used to create the maternal health risk prediction module. Random Forest model had an accuracy of 93.08% and was ultimately adopted as the final model because of its ability to generate good prediction accuracy and its suitability to be used in real time. The optimized Random Forest model was embedded into a web application using Flask for real-time maternal health risk prediction. The optimized Random Forest model was encapsulated in a web application using Flask for real-time maternal health risk prediction.

A YOLO based object detection method is implemented to develop a human fall detection module. Four models: YOLOv8n, YOLOv9t, YOLO11s, and YOLO26n were trained and tested using a dataset of Fall and Non-Fall classes. Precision, recall, mAP@50, mAP@50–95 and inference time were used to compare the models. YOLOv8n was chosen as the final model due to its high recall of 99.7%, and the minimum inference time of 2.3 ms per image, which shows it is suitable for real-time fall detection.

Finally, the chosen models were deployed for real-time use using Flask for maternal health risk prediction and OpenCV for camera integration for fall detection. The outcome shows that the developed AI modules are able to support continuous monitoring of a mother's health, identifying and determining the health risk level, while also detecting possible fall events in real-time.

Chapter 5

Results and Evaluation

5.1 Experimental Setup

The Materna AI system was built using an ESP32 microcontroller, sensors, a Flask-based backend and a MongoDB database. The system has ECG monitoring, fall detection, GPS tracking and doctor consultation abilities. Sensor data was sent via Wi-Fi. Shown on a web-based patient dashboard. The system was tested with functional test cases including patient login, ECG monitoring, fall detection, GPS tracking, doctor consultation, message exchange and alert creation. The expected results were matched with the findings to check how well each part worked.

5.2 Hardware Implementation

The health monitoring system that was developed was put together with an ESP32 microcontroller and a number of sensing and communication modules. The health monitoring system's parts were wired on a breadboard with jumper wires to allow testing and fast prototyping.

The health monitoring system contains a MAX30102/MAX30105 sensor that measures heart rate and blood oxygen saturation an ECG sensor that records the heart's signal an MPU6050 sensor that tracks motion and detects falls and a BMP280 sensor that measures temperature and pressure. A GPS module was added so the health monitoring system could find the patient's location and an HC-05 Bluetooth module was added to support wireless communication.

The ESP32 serves as the controller for the health monitoring system. The ESP32 gathers data from the sensors processes the readings shows the information, on the OLED display and sends the data to the health monitoring system via Wi-Fi and Bluetooth.

The complete hardware prototype of the health monitoring system is shown in Figure 5.1.

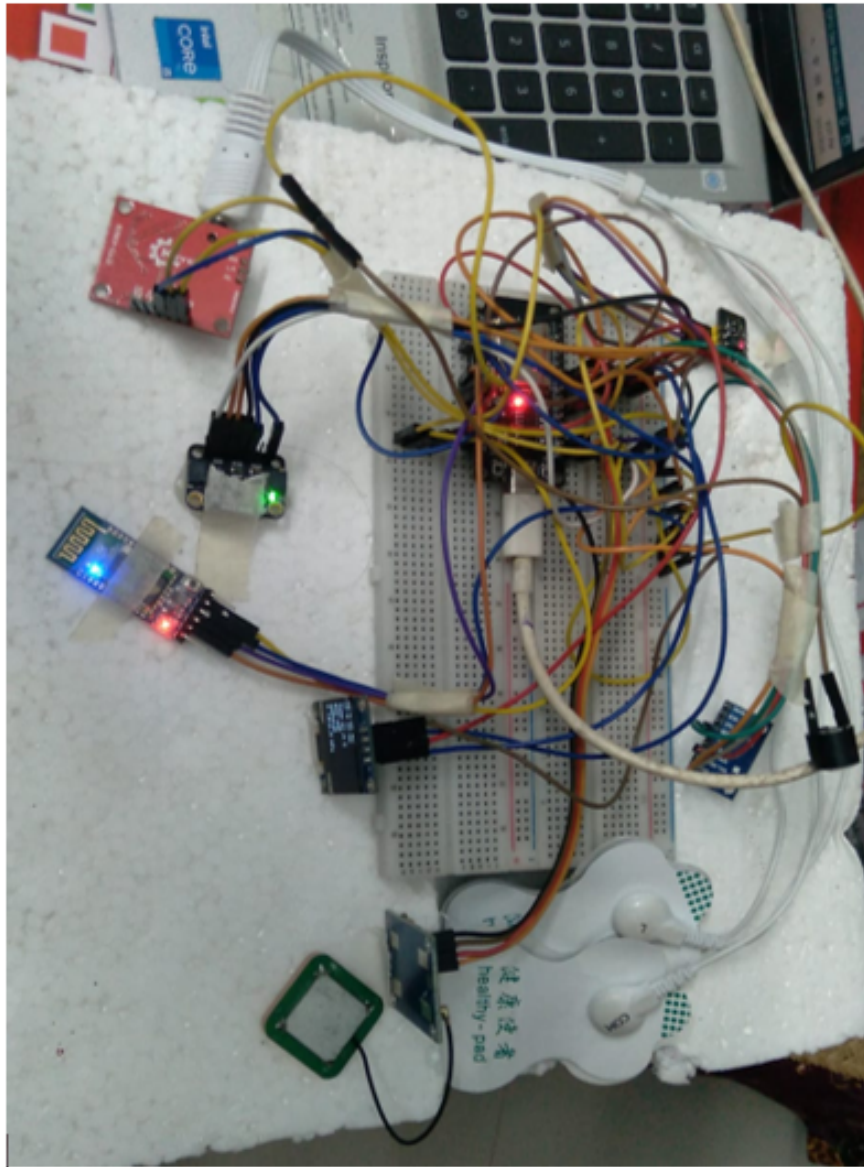


Figure 5.1: Assembled hardware prototype of the proposed health monitoring system.

5.3 Bluetooth Communication Results

The bluetooth communication using an HC-05 Bluetooth module and a terminal application, on a smartphone. The ESP32 successfully connected to the HC-05 module. Transmitted ECG data wirelessly. I watched the ECG data stream come through the terminal without any delay.

The received data included the ECG value and a Google Maps link generated from the GPS coordinates. The ECG readings were displayed in the terminal in time confirming successful Bluetooth data transmission and GPS location sharing.

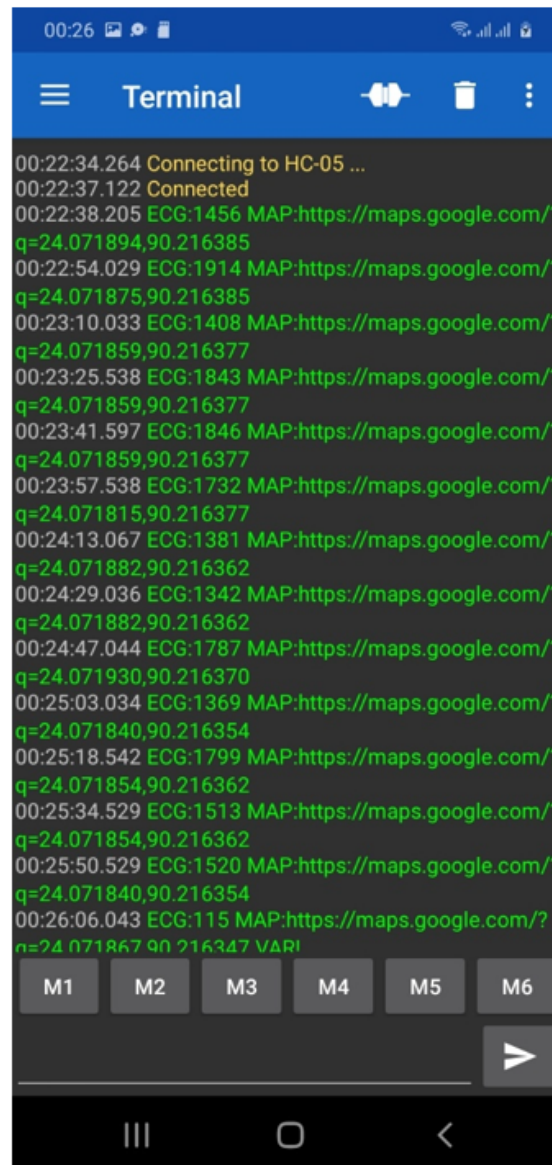


Figure 5.2: Bluetooth terminal showing ECG readings and Google Maps location links received from the ESP32.

5.4 YOLO based Fall Detection System

The fall detection system uses a webcam to watch the patient's movements. The user can stop the fall detection system using the provided buttons. The fall detection system shows the current movement status, such, as normal movement. Helps identify possible fall incidents.

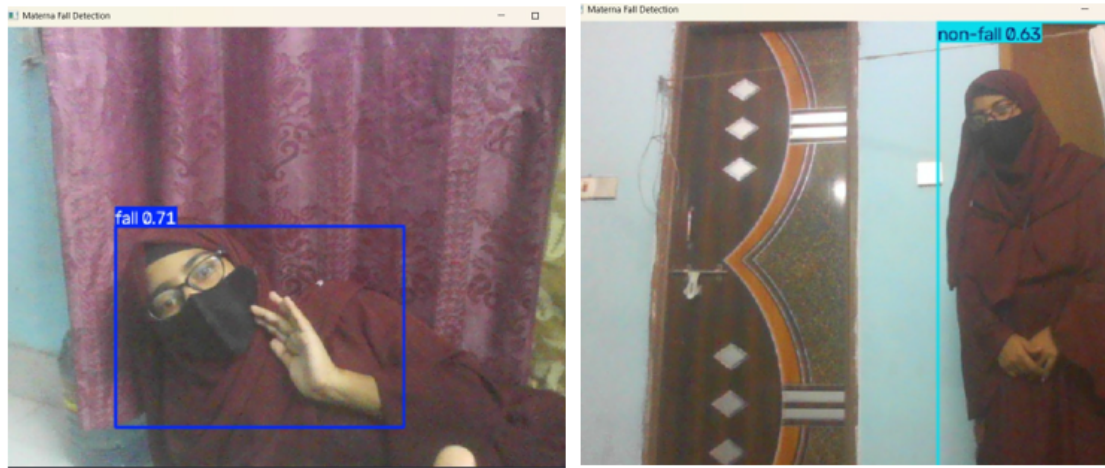


Figure 5.3: Webcam-based fall detection interface showing fall and non-fall status

5.5 Discussion

The Materna AI system brings together sensors, machine learning, fall detection, GPS, Bluetooth and web-based monitoring to support real-time maternal healthcare. The ESP32 gathers data from the sensors and shows the results on the OLED screen.

Random Forest gave the accuracy at 93.08 percent for predicting maternal health risks. For detecting falls YOLOv8n achieved 99.7 percent recall, 99.4 percent mAP@50 and an inference time of 2.3 milliseconds making it well-suited for real-time use.

The ECG-based buzzer sounds an alert when the raw ECG value drops below 500 or rises above 3000. Bluetooth allows short-range data transfer while Wi-Fi and Flask enable monitoring. GPS helps track the patient's location in case of emergencies.

Still there are challenges such as sensor noise, reliance on network connections limitations, with GPS accuracy and the use of prototype-level ECG thresholds. More clinical testing and better calibration are needed before this system can be used in world medical settings.

Chapter 6

Conclusion and Future Work

The Materna AI system successfully integrates sensors, machine learning, fall detection, GPS, Bluetooth and web-based monitoring for maternal health. The ESP32 Processes health data and shows the results on an OLED screen. Random Forest gives predictions for health risk while YOLO-based fall detection and an ECG-based buzzer provide extra safety. Wi-Fi, Flask, ThingSpeak and GPS help, with monitoring and emergency location tracking. Overall the proposed system gives an scalable prototype for real-time maternal health monitoring but more clinical validation is still needed.

6.1 Summary of Contributions

The Materna AI system offers an smart solution for monitoring maternal health in real time. It brings together technologies like IoT sensors, machine learning, fall detection, GPS tracking, Bluetooth communication and web-based monitoring to make patient care safer and more accessible. The system gathers health information using an ESP32-based controller and shows the data right on a local OLED display. It also includes ECG-based buzzer alerts that sound when readings are not normal. Data can be sent remotely through Wi-Fi uploaded to ThingSpeak.

In addition the developed dashboard lets doctors keep track of health check personal details and medical history conduct online consultations and follow the patient's location using GPS. The machine learning module helps predict risks during pregnancy. Altogether this system supports round-the-clock monitoring, quick emergency help and stronger communication, between patients and healthcare staff.

6.2 Key Findings and Performance Metrics

The Materna AI system works well by combining sensors, machine learning and web tools for maternal healthcare. The main results and how well it works are listed below.

The Random Forest model had an accuracy of 93.08 percent when predicting risks for health. The system divides patients into medium and high risk based on their health data.

For detecting images YOLOv8n had 99.7 percent recall and 99.4 percent mAP@50 with a time of 2.3 milliseconds. These results show that the model can find visual issues quickly.

The ESP32 hardware system collects data from the ECG MAX30100, MPU6050, BMP280 and GPS sensors. The OLED screen shows current local information and the ECG-based buzzer sounds a warning when the ECG value is under 500 or, over 3000.

Bluetooth sends ECG data and GPS information. Wi-Fi sends sensor data to the Flask-based dashboard and the ThingSpeak platform. The dashboard lets doctors check details, health data, medical history, appointments and GPS location from a distance.

Overall the system shows that using IoT devices, machine learning and remote monitoring can help keep track of health and provide emergency help.

6.3 Limitations and Future Scope

The Materna AI system has reached its goals but the Materna AI system still has some limits and that gives chances, for future improvement.

1. **Sensor Accuracy and Calibration:** The health data that is collected can be influenced by sensor noise, placement of the device and changes, in the environment. In the future efforts should be made to improve calibration use signal filtering methods and rely on sensors that have been clinically validated.
2. **Real-Time Emergency Response:** The current system mainly provides alerts through the buzzer, Bluetooth and the web dashboard. The system could be improved by adding call notifications, to family members, doctors or emergency services so that faster medical assistance can be delivered.
3. **Security and Privacy:** Patient health information sent through Wi-Fi, Bluetooth

and the web dashboard could be at risk, for security breaches and privacy issues. To protect data future efforts should focus on data encryption secure authentication methods, access control systems and safe cloud storage solutions. These steps are important to keep confidentiality strong and reliable.

Finally Materna AI offers a platform for keeping track of a mothers health all the time. It uses sensors, GPS, Bluetooth and a web-based dashboard. More improvements can make the system better at giving results being dependable keeping information safe and being ready, for real-life medical use.

Bibliography

- [1] S. Tappari, N. Sreshta, M. S. Aare, *et al.*, “A study on maternal and child health monitoring system using iot and machine learning,” in *Proc. IEEE Int. Conf. on Emerging Computing and Intelligent Technologies (ICoECIT)*, 2026.
- [2] K. S. Karventhan, D. Marshiana, and K. P. Indira, “Development of an iot-driven wearable belt for comprehensive maternal and fetal health tracking,” in *Proc. Int. Conf. on Electronics and Renewable Systems (ICEARS)*, 2026.
- [3] A. S. Banga, “Enhancing cybersecurity in iot-based maternal health monitoring systems using machine learning algorithms,” *Islamic University Journal of Applied Sciences*, vol. 8, no. 1, pp. 78–109, 2026.
- [4] A. T. Abu-Jassar, H. Attar, A. Amer, *et al.*, “Remote monitoring system of patient status in social iot environments using amazon web services technologies and smart health care,” *International Journal of Crowd Science*, vol. 9, no. 2, pp. 110–125, 2025.
- [5] I. Seflek and M. Barstuğan, “Radar-based fall detection using micro-doppler signatures: A comparative analysis of yolo architectures,” *Sensors*, vol. 26, no. 9, p. 2650, 2026.
- [6] M. E. N. Gomes, P. S. G. de Mattos-Neto, C. Zanchettin, L. N. Matos, and B. S. Morais, “Fall detection based on computer vision: A systematic mapping study,” *IEEE Journal of Biomedical and Health Informatics*, 2026.

- [7] F. Harrou, N. Zerrouki, Y. Sun, and A. Houacine, "Vision-based fall detection system for improving safety of elderly people," *IEEE Instrumentation & Measurement Magazine*, pp. 49–55, 2017.
- [8] G. Yenurkar, V. Khangar, J. Shinde, *et al.*, "Comprehensive review on iot architectures for maternal and fetal health monitoring in high-risk pregnancies," in *Proc. 7th Int. Conf. on Artificial Intelligence and Speech Technology (AIST)*, 2025.
- [9] M. Ahmed and M. A. Kashem, "Iot based risk level prediction model for maternal health care in the context of bangladesh," in *Proc. 2nd Int. Conf. on Sustainable Technologies for Industry 4.0 (STI)*, (Dhaka, Bangladesh), pp. 1–6, December 2020.
- [10] Y.-J. Hu, P.-C. Hou, H.-Y. Chen, T.-F. Hou, and C.-H. Yen, "A low-cost noncontact vital sign monitoring and fall detection system using mmwave radar and yolo-based edge artificial intelligence," in *2026 IEEE International Conference on Consumer Electronics–Taiwan (ICCE-Taiwan)*, 2026.
- [11] L. Liu, Y. Pu, J. Fan, Y. Yan, W. Liu, K. Luo, Y. Wang, G. Zhao, T. Chen, P. D. Puiu, and H. Huang, "Wearable sensors, data processing, and artificial intelligence in pregnancy monitoring: A review," *Sensors*, vol. 24, no. 19, p. 6426, 2024.
- [12] G. Milan, L. Ariniello, K. Baca-Motes, A. Faksh, J. K. Kueper, J. A. Pandit, T. Ajayi, and G. Quer, "Association between wearable sensor signals and expected hormonal changes in pregnancy," *eBioMedicine*, vol. 119, p. 105888, 2025.
- [13] A. Malde, V. G. Prabhu, D. Banga, M. Hsieh, C. Renduchintala, and R. Pirrallo, "A machine learning approach for predicting maternal health risks in lower-middle-income countries using sparse data and vital signs," *Future Internet*, vol. 17, no. 5, p. 190, 2025.
- [14] N. Spicher, J. M. N. D'Souza, A. R. K. Williamson, *et al.*, "Comparative analysis of machine learning approaches for fetal movement detection using wearable sensors," *Scientific Reports*, vol. 15, p. 11234, 2025.

- [15] B.-C. Bernad, M.-C. Tomescu, D. E. Velimirovici, M. Andor, D. Lungeanu, V. Enatescu, A.-I. Bucur, A. Lascu, A.-L. Rata, E. S. Bernad, *et al.*, “Impact of stress and anxiety on cardiovascular health in pregnancy: A scoping review,” *Journal of Clinical Medicine*, vol. 14, no. 3, p. 909, 2025.
- [16] S. G. Ahmad, M. A. Arif, A. Hassan, K. Ayyub, E. U. Munir, and N. Ramzan, “Iot-based smart wearable belt for tracking fetal kicks and movements in expectant mothers,” *IEEE Sensors Journal*, vol. 25, pp. 27322–27333, July 2025.
- [17] B. Priyanka, V. M. Kalaivanan, R. A. Pavish, and M. Kanageshwaran, “Iot based pregnancy women health monitoring system for prenatal care,” in *Proc. 7th International Conference on Advanced Computing and Communication Systems (ICACCS)*, pp. 1264–1269, 2021.
- [18] R. Ettiyan and V. Geetha, “Iod-nets – an iot based intelligent health care monitoring system for ambulatory pregnant mothers and fetuses,” *Measurement: Sensors*, vol. 27, p. 100781, 2023.
- [19] H. Allahem and S. Sampalli, “Automated labour detection framework to monitor pregnant women with a high risk of premature labour using machine learning and deep learning,” *Informatics in Medicine Unlocked*, vol. 28, p. 100771, 2022.
- [20] S. L. Kailan, W. H. M. Kurdi, A. H. Najim, and M. N. Kadhim, “Efficient ecg classification based on machine learning and feature selection algorithm for iot-5g enabled health monitoring systems,” *International Journal of Intelligent Engineering and Systems*, vol. 18, no. 1, pp. 1187–1199, 2025.
- [21] J. V. Suman, S. K. Velagaleti, R. Khusniddin, A. Ashirova, B. Sivakumar, and P. RamaMohan, “Low-power wireless health monitoring system for smart hospital applications,” in *Proceedings of the International Conference on Electronics and Renewable Systems (ICEARS)*, 2026.
- [22] X. Qu, X. Ding, X. Li, W. Yu, and Y.-L. Xu, “Assessing and mitigating impacts of structural inclination on acceleration measurements and estimated displacements

- using integrated gnss and accelerometer structural health monitoring systems,” *Measurement*, vol. 253, p. 117578, 2025.
- [23] K. Qi, “Advancing hospital healthcare: Achieving iot-based secure health monitoring through multilayer machine learning,” *Journal of Big Data*, vol. 12, no. 1, 2025.
- [24] G. L. Simegn and M. Z. Degu, “Maternal health risk analysis, automated pregnancy risk level identification and status monitoring system based on multivariable clinical data,” *Health Technology*, vol. 9, 2025.
- [25] R. M. S. M. Domingues, A. S. Rodrigues, M. A. B. Dias, V. Saraceni, R. P. V. Francisco, R. S. Pinheiro, and C. M. Coeli, “Maternal health surveillance panel: A tool for expanding epidemiological surveillance of women’s health and its determinants,” *Revista Brasileira de Epidemiologia*, vol. 27, 2024.
- [26] World Health Organization, United Nations Children’s Fund, United Nations Population Fund, World Bank Group, and United Nations Department of Economic and Social Affairs, “Trends in maternal mortality 2000 to 2020: Estimates by who, unicef, unfpa, world bank group and undesa/population division,” tech. rep., World Health Organization, Geneva, Switzerland, 2023.
- [27] H. N. G. and A. S. S., “Study exploring the use of iot and ml in pregnant women healthcare monitoring system,” in *Proceedings of the 2nd International Conference on Advances in Information Technology (ICAIT)*, 2024.
- [28] L. Say, D. Chou, A. Gemmill, Ö. Tunçalp, A.-B. Moller, J. Daniels, A. M. Gülmezoglu, M. Temmerman, and L. Alkema, “Global causes of maternal death: a who systematic analysis,” *The Lancet Global Health*, vol. 2, no. 6, pp. e323–e333, 2014.
- [29] M. S. H. Talpur, M. H. Shaikh, R. A. Buriro, H. S. Talpur, F. Talpur, H. Shafi, and M. A. Shaikh, “Internet of things as intimating for pregnant women’s healthcare: An impending privacy issues,” *TELKOMNIKA (Telecommunication, Computing, Electronics and Control)*, vol. 12, no. 6, 2014.

- [30] V. Santhi, K. Ramya, A. P. J. Tarana, and G. Vinitha, “Iot based wearable health monitoring system for pregnant ladies using cc3200,” *International Journal of Advanced Research Methodology in Engineering & Technology*, vol. 1, pp. 56–60, May 2017.
- [31] R. Ettiyan and V. Geetha, “A survey of health care monitoring system for maternity women using internet-of-things,” in *Proceedings of the 3rd International Conference on Intelligent Sustainable Systems (ICISS)*, pp. 1290–1296, 2020.
- [32] J. A. L. Marques, T. Han, W. Wu, J. P. d. V. Madeiro, A. V. Lira Neto, R. Gravina, G. Fortino, and V. H. C. de Albuquerque, “Iot-based smart health system for ambulatory maternal and fetal monitoring,” *IEEE Internet of Things Journal*, vol. 8, pp. 16814–16824, Dec. 2021.
- [33] M. Assaduzzaman, A. A. Mamun, and M. Z. Hasan, “Early prediction of maternal health risk factors using machine learning techniques,” in *2023 International Conference for Advancement in Technology (ICONAT)*, (Goa, India), 2023.
- [34] L. Spicher, C. Bell, X. Huan, and K. Sienko, “Exploring random forest machine learning for fetal movement detection using abdominal acceleration and angular rate data,” in *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2024.
- [35] S. A. Madhi, L. M. Rivera, X. Sáez-Llorens, C. Menéndez, N. Carrim-Ganey, M. F. Cotton, D. Katzman, M. M. Luttig, R. Candelario, S. Baker, and M. Roychoudhury, “Factors influencing access of pregnant women and their infants to their local health-care system: a prospective, multi-centre, observational study,” *BMC Pregnancy and Childbirth*, vol. 18, no. 29, 2018.
- [36] K. J. Marwa, A. Njalika, D. Ruganuzza, D. Katabalo, and E. Kamugisha, “Self-medication among pregnant women attending antenatal clinic at makongoro health centre in mwanza, tanzania: a challenge to health systems,” *BMC Pregnancy and Childbirth*, vol. 18, no. 16, 2018.

- [37] M. M. Hossain, M. A. Kashem, N. M. Nayan, and M. A. Chowdhury, “A medical cyber-physical system for predicting maternal health in developing countries using machine learning,” *Healthcare Analytics*, vol. 5, p. 100285, 2024.
- [38] R. Escuriet, J. White, K. Beeckman, L. Frith, F. Leon-Larios, C. Loytved, A. Luyben, M. Sinclair, E. van Teijlingen, and E. C. A. ISO907, “Assessing the performance of maternity care in europe: a critical exploration of tools and indicators,” *BMC Health Services Research*, vol. 15, no. 491, 2015.
- [39] Y. Gupta, S. Kumar, and V. Mago, “Pregnancy health monitoring system based on biosignal analysis,” in *2019 42nd International Conference on Telecommunications and Signal Processing (TSP)*, (Budapest, Hungary), pp. 664–667, 2019.
- [40] M. Ramla, S. Sangeetha, and S. Nickolas, “Fetal health state monitoring using decision tree classifier from cardiotocography measurements,” in *Proceedings of the Second International Conference on Intelligent Computing and Control Systems (ICICCS)*, pp. 1799–1803, 2018.
- [41] S. Sharma, H. Sidhu, and S. Kaur, “Analytical study of intrauterine fetal death cases and associated maternal conditions,” *International Journal of Applied and Basic Medical Research*, vol. 6, no. 1, pp. 11–13, 2016.
- [42] S. Kumar, Y. Gupta, and V. Mago, “Health-monitoring of pregnant women: Design requirements, and proposed reference architecture,” in *2019 16th IEEE Annual Consumer Communications & Networking Conference (CCNC)*, 2019.
- [43] F. J. Dian, R. Vahidnia, and A. Rahmati, “Wearables and the internet of things (iot), applications, opportunities, and challenges: A survey,” *IEEE Access*, vol. 8, pp. 69200–69211, 2020.
- [44] S. Majumder, T. Mondal, and M. J. Deen, “Wearable sensors for remote health monitoring,” *Sensors*, vol. 17, no. 1, p. 130, 2017.

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